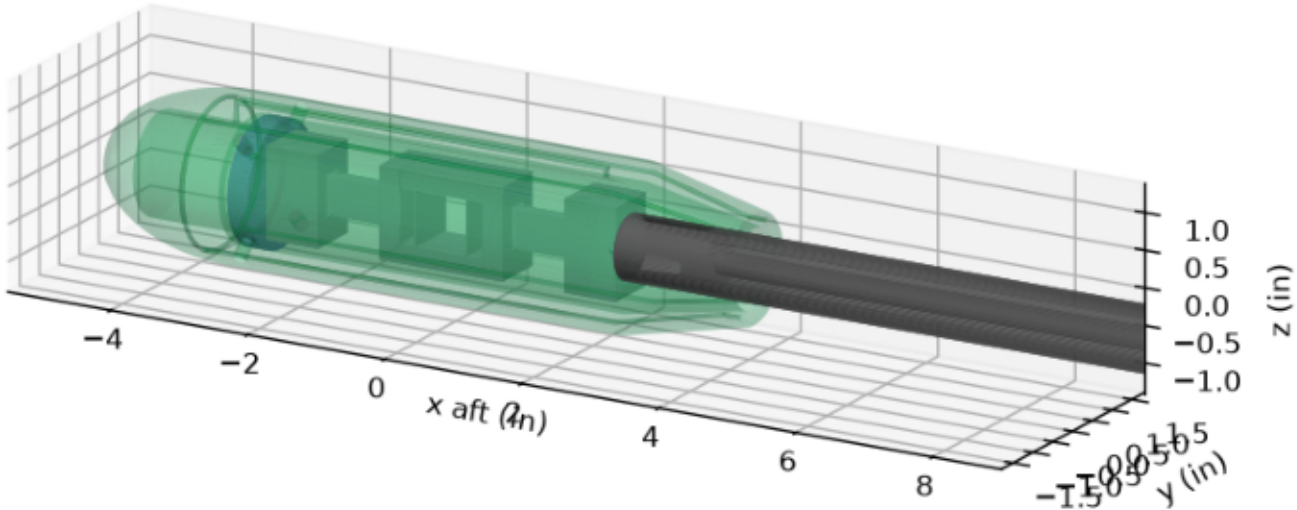


Flight assembly: instrumented 2.25 in pod (transparent), puck, 6-component balance, sting



NOTES:

1. GAUGE AND WIRE THE BALANCE (24 GAUGES, 6 BRIDGES) BEFORE ASSEMBLY; SEE THE GAUGE LAYOUT DRAWING.
2. PUCK TO BALANCE FRONT FACE: 2 DOWELS + 2 SCREWS. STING ON THE SPIGOT: CROSS PIN + 2 SET SCREWS.
3. WIRES RUN AFT THROUGH THE BALANCE AND THE STING BORE TO J0 AT THE ROLL HOUSING.
4. THE MODEL POD (WT-6C-004) IS 3D PRINTED FROM ITS STEP FILE; THE SHEET GIVES THE FIT AND INSPECTION CHECKS.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

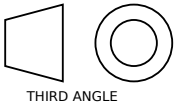
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

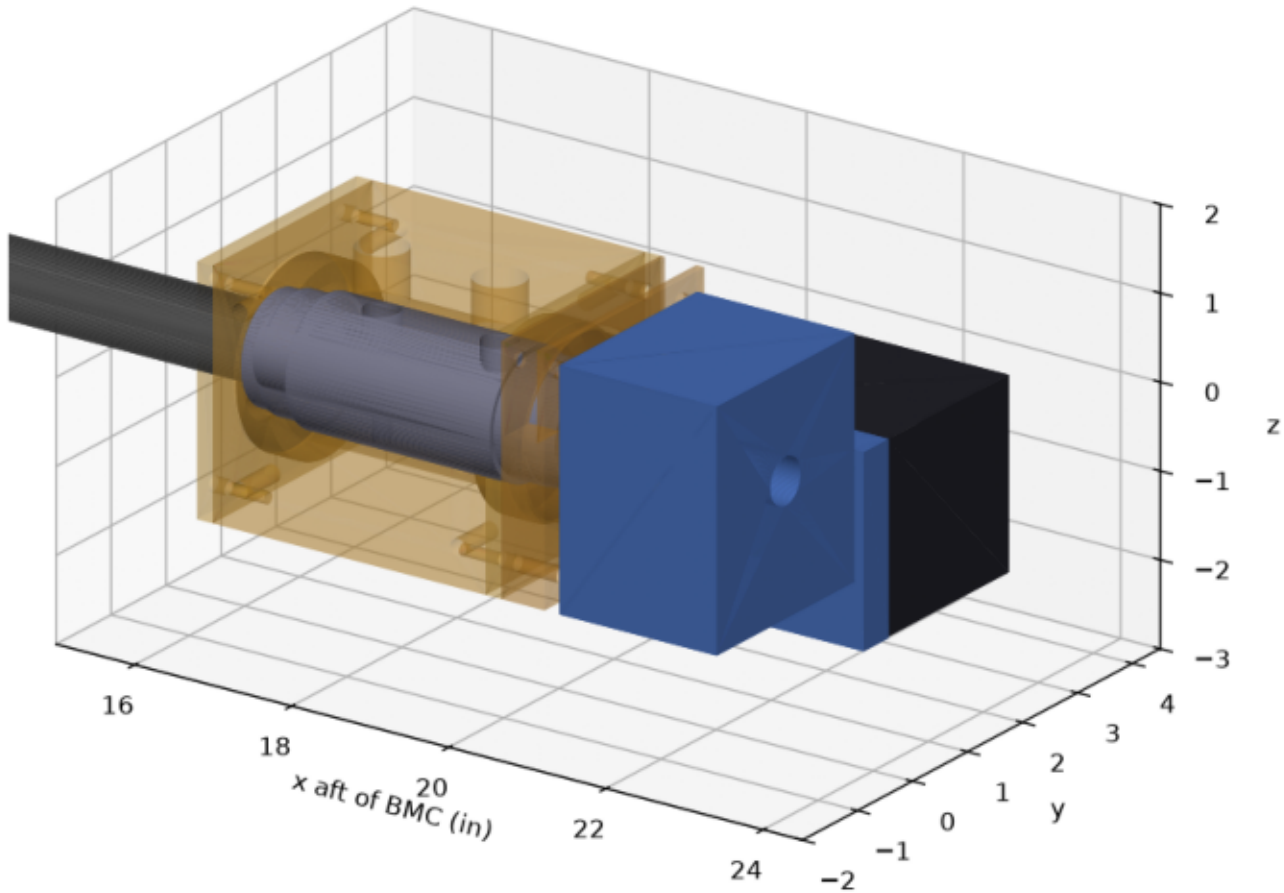
ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-001	Balance, 6-component (gauged)	1	7075-T651
2	WT-6C-002	Puck, model interface	1	6061-T6
3	WT-6C-003	Sting	1	4140 28-32 HRC
4	WT-6C-004	Pod, reference model, 2.25 in	1	PC-ABS, 3D printed
5	-	Strain gauge 350 ohm, CEA-13 class	24	Vishay / bought
6	-	Dowel pin 1/8 x 1/2	2	bought
7	-	SHCS #10-32 x 5/8 (puck)	2	bought
8	-	Cross pin 3/16 dowel (sting)	1	bought
9	-	Set screw 1/4-28 cup point	2	bought
10	-	SHCS #8-32 x 3/8 (pod)	4	bought
11	-	Wire 32 AWG PTFE + J0 DC-37	1	see Electronics

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
ASSEMBLY, FLIGHT (BALANCE + STING + MODEL)							
MATERIAL				FINISH			
SEE PARTS LIST				SEE PART DRAWINGS			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-100		A	B	NTS		-	1 OF 36

ROLL SPINDLE AT ROLL 0 (CAD RENDER, NTS)

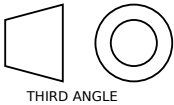
Roll spindle at roll +0 deg: housing/caps/adapter transparent, NMRV030 50:1 + NEMA 23, AS5048A encoder



NOTES:

- 1. PRESS BOTH BEARINGS ONTO THE SPINDLE SEATS; FIT INTO THE HOUSING; CLAMP WITH THE CAPS. SHIM THE REAR CAP FOR .002-.004 AXIAL FLOAT; TIGHTEN THE KM7 LOCKNUT.
- 2. STING FLATS IN THE D-D BORE; 2 BOLTS THROUGH THE TOP WINDOWS AT ROLL 0.
- 3. AS5048A CHIP ON THE SPINDLE AXIS WITHIN 0.25 MM, 1.5 MM AIR GAP.
- 4. BALANCE CABLE EXITS AT X = 20.53: LEAVE A SERVICE LOOP FOR 270 DEG OF TWIST (CALIBRATION MODE).

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X ±.03 .XX ±.010
.XXX ±.005 .XXXX ±.0005
ANGLES ±0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 µin Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

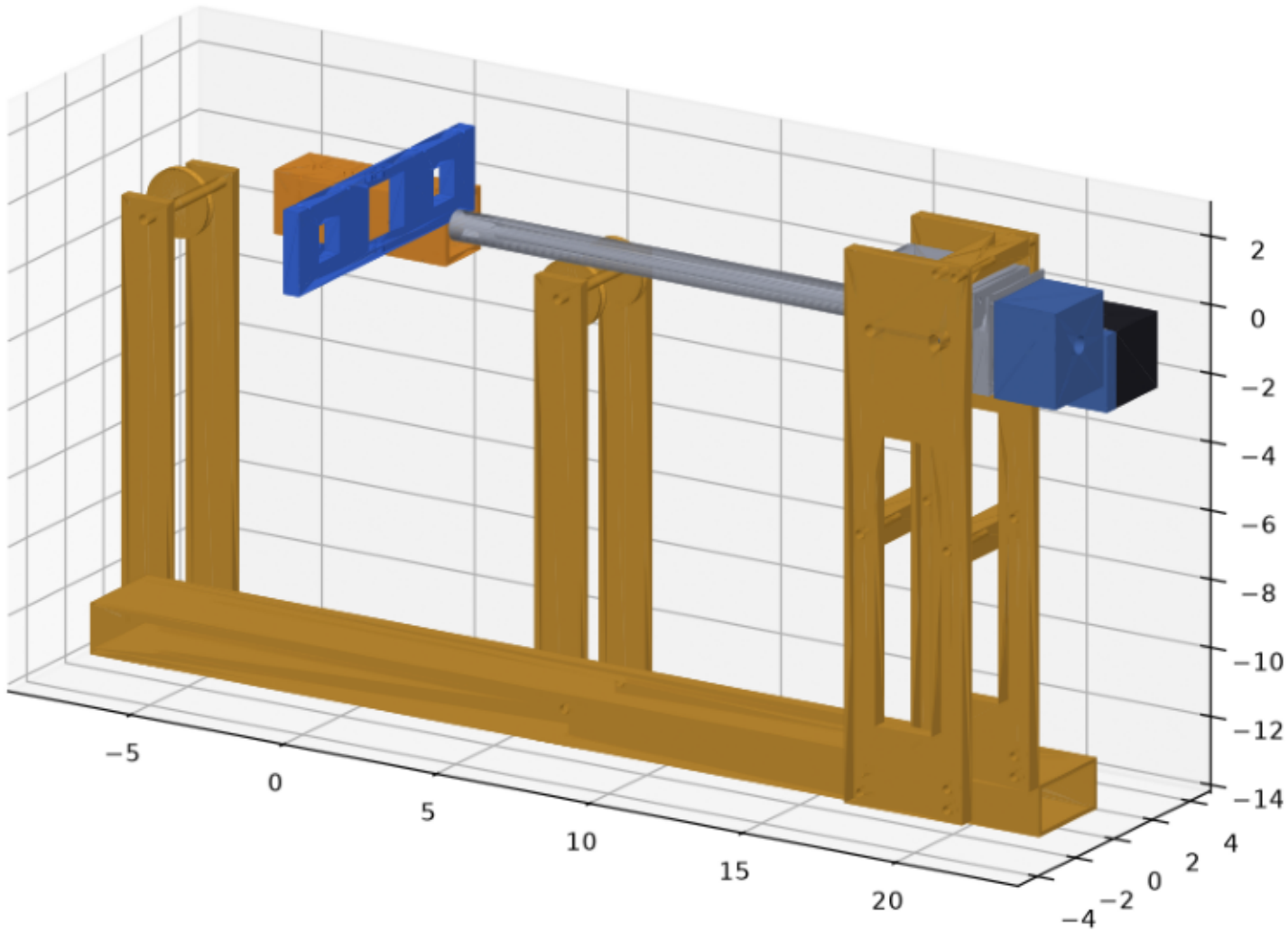
ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-201	Spindle, roll	1	4140
2	WT-6C-202	Housing, roll spindle	1	6061-T6
3	WT-6C-203-1	Cap, bearing, front	1	6061-T6
4	WT-6C-203-2	Cap, bearing, rear	1	6061-T6
5	WT-6C-204	Adapter, gearbox	1	6061-T6
6	WT-6C-205	Bracket, roll encoder	1	6061-T6
7	-	Bearing 6907-2RS (35x55x10)	2	bought
8	-	Locknut KM7 + lock washer MB7	1	bought
9	-	Worm gearbox NMRV030 50:1, NEMA 23 input	1	bought
10	-	Stepper motor NEMA 23	1	bought
11	-	Encoder AS5048A board + 6x2.5 diametric magnet	1	bought
12	-	SHCS #8-32 (caps, adapter)	16	bought
13	-	SHCS M6 (gearbox), M3 standoffs (encoder)	1	bought
14	-	Shims .002/.005 (rear cap float)	1	bought

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, ROLL SPINDLE								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-200		A	B	NTS		-		2 OF 36

CALIBRATION ON THE ROLL SPINDLE (CAD RENDER, NTS)

6-component calibration on the roll spindle: square sleeve, roll arm, yoke at x = -2, drag/thrust pulleys



NOTES:

1. THE BALANCE IS CALIBRATED ON THE REAL ROLL SPINDLE: THE HOUSING PIVOTS ON ITS FRONT STUD BETWEEN THE HEAD PLATES; THE REAR STUD SITS IN THE LEVELING SLOT; RE-LEVEL WITH THE JACK SCREW AFTER EVERY LOAD.
2. SLEEVE REPLACES THE PUCK. YOKE AT ROLL 0/90/180/270, STIRRUP AT 45/135/315, ROLL ARM AT 0/180.
3. BENCH MODE ROLL RANGE -90..+180 DEG. SCHEDULE AND FIT: analysis/balance_calibration_fit_6c.py.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

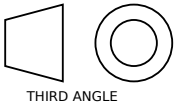
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



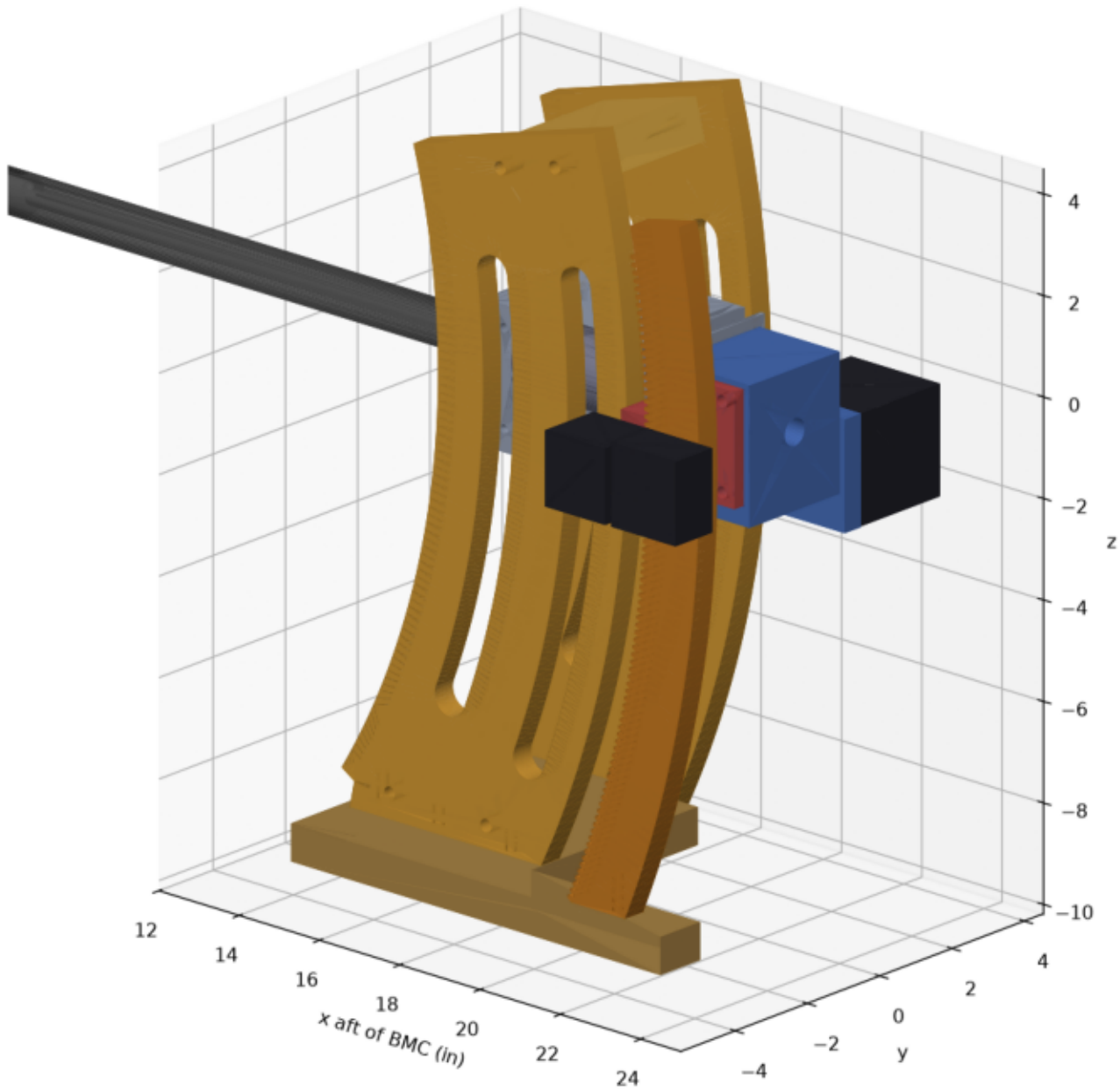
PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-301	Sleeve, calibration	1	6061-T6
2	WT-6C-302	Arm, roll calibration	1	6061-T6
3	WT-6C-303	Stirrup, 45 deg	1	6061-T6
4	WT-6C-304	Yoke, calibration	1	6061-T6
5	WT-6C-305	Plate, head	2	6061-T6
6	WT-6C-306-1	Bridge, jack	1	6061-T6
7	WT-6C-306-2	Spacer, cross	2	6061-T6
8	WT-6C-307	Tube, base	1	6061-T6 tube
9	WT-6C-308	Upright, pulley	4	6061-T6
10	WT-6C-309	Axle, pulley	2	416 SS ground
11	WT-6C-310	Pulley, cable, 2 in	3	6061-T6
12	-	Bearing R5-2Z (5/16 x 1/2)	3	bought
13	-	Shaft collar 5/16	4	bought
14	WT-6C-200	Roll spindle assy (with balance + sting)	1	see WT-6C-200
15	-	Stud 3/8 plain + nut (housing pivot/level)	4	bought
16	-	Jack screw 1/4-28 brass tip	1	bought
17	-	Ground pin 1/4 (load pin), dowels 1/4	3	bought
18	-	Cable 1/16 SS, eyebolt 1/4-20, hooks	1	bought
19	-	Dead weights, calibrated (Class F)	1	bought
20	-	SHCS 1/4-20 + nuts	16	bought

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, 6-COMPONENT CALIBRATION								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-300		A	B	NTS		-		3 OF 36

PITCH SECTOR WITH ROLL SPINDLE (CAD RENDER, NTS)

Motorized pitch sector at alpha 0: braced cheeks with arc tracks, rack plate + pinion drive (-y side), roll spindle carried on 4 cam followers



NOTES:

1. BOLT THE CHEEKS, TIE AND SPACERS TOGETHER ON THE BASE PLATE FIRST; CHECK THE CHEEK SPACING (3.000).
2. INSTALL THE CAM FOLLOWERS IN THE ROLL HOUSING; LOWER THE ROLL ASSEMBLY INTO THE ARC SLOTS.
3. SET THE RACK-PINION BACKLASH .004-.008 IN; GRAVITY KEEPS THE SAME FLANK LOADED.
4. RANGE -6..+20 DEG; PITCH RATE LIMITED TO 2 DEG/S.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

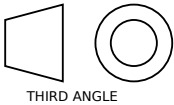
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

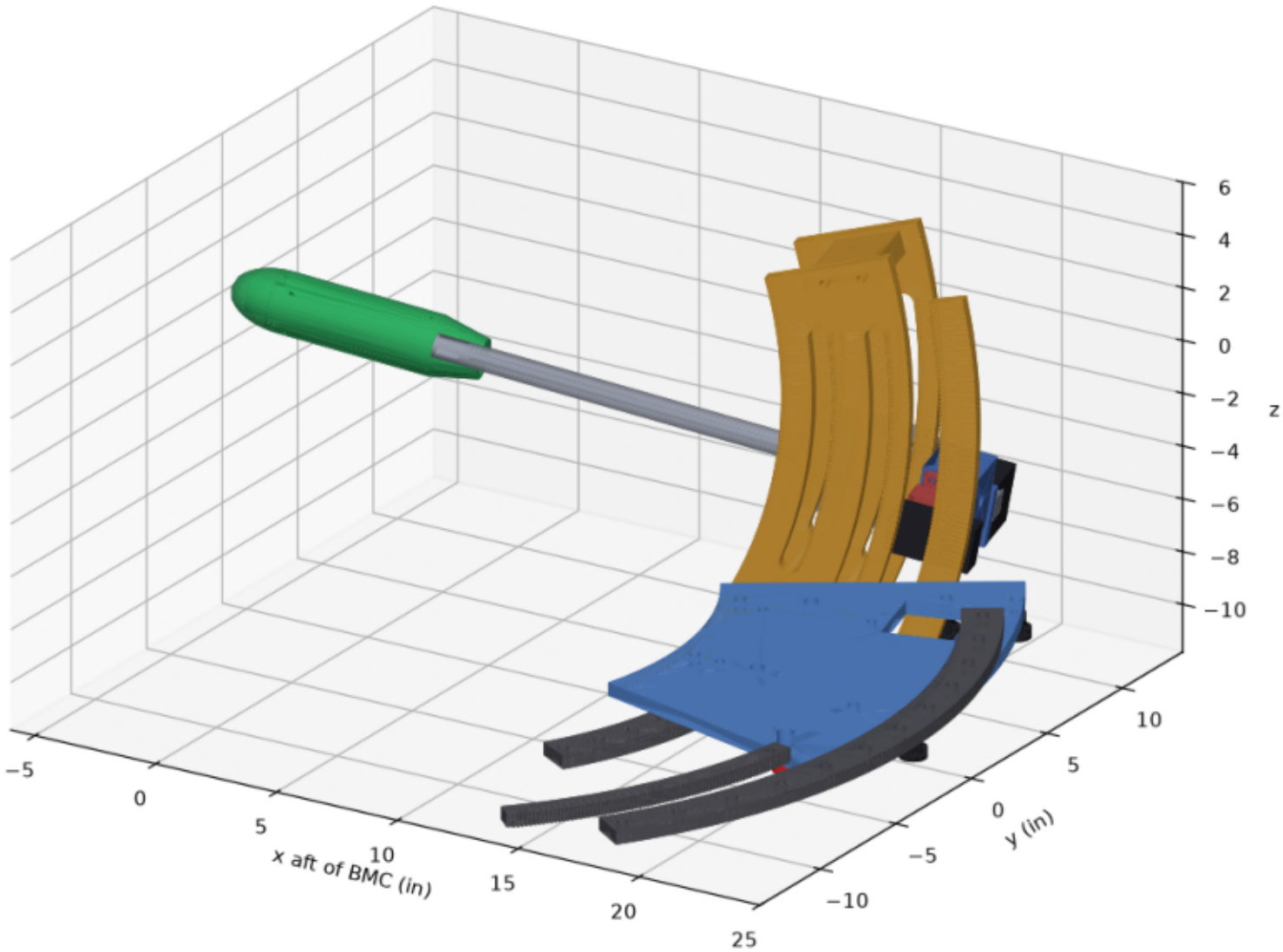
ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-401	Cheek, pitch sector	2	6061-T6
2	WT-6C-402	Rack plate, pitch	1	6061-T6
3	WT-6C-403	Plate, base, pitch	1	6061-T6
4	WT-6C-404-1	Tie, top	1	6061-T6
5	WT-6C-404-2/3	Spacers, fwd / aft	2	6061-T6
6	WT-6C-405	Bracket, pitch drive	1	6061-T6
7	WT-6C-406	Pinion shaft (+ stock m1.25 20T gear)	1	4140 + bought gear
8	-	Bearing 608-2RS	2	bought
9	-	Circlip DIN 471-8	1	bought
10	-	Cam follower 3/4 in, 3/8-24 stud	4	bought
11	-	Worm gearmotor 40:1 + NEMA 17	1	bought
12	WT-6C-200	Roll spindle assy	1	see WT-6C-200
13	WT-6C-100	Flight assy (balance, sting, model)	1	see WT-6C-100
14	-	SHCS 1/4-20 (cheeks, tie, spacers, rack)	14	bought
15	-	SHCS M6 (drive bracket)	4	bought
16	-	Limit switches + inclinometer	1	see Electronics

NEXT ASSY: WT-6C-500 YAW CARRIAGE ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, MOTORIZED PITCH SECTOR								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-400		A	B	NTS		-		4 OF 36

COMPLETE MOTION SYSTEM ON THE YAW CARRIAGE (CAD RENDER, NTS)

Complete motion system at yaw +15, pitch +10: floor rails -> yaw carriage -> pitch sector -> roll spindle -> sting -> model



NOTES:

1. PRELIMINARY: RAIL POSITIONS AND FLOOR HOLES DEPEND ON THE DOWNSTREAM BAY FLOOR (TO CONFIRM).
2. INSTALL THE RAILS CONCENTRIC WITH THE YAW AXIS (THROUGH THE BMC) USING A RADIUS BAR ON A PIN.
3. PRELOAD EACH TRUCK WITH ITS ECCENTRIC SPACER UNTIL THE WHEELS TURN WITH LIGHT DRAG.
4. RANGE +-15 DEG YAW; THE PLATE REACHES ABOUT 2.3 IN UNDER THE TEST-SECTION EXIT AT FULL YAW.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

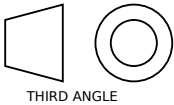
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL

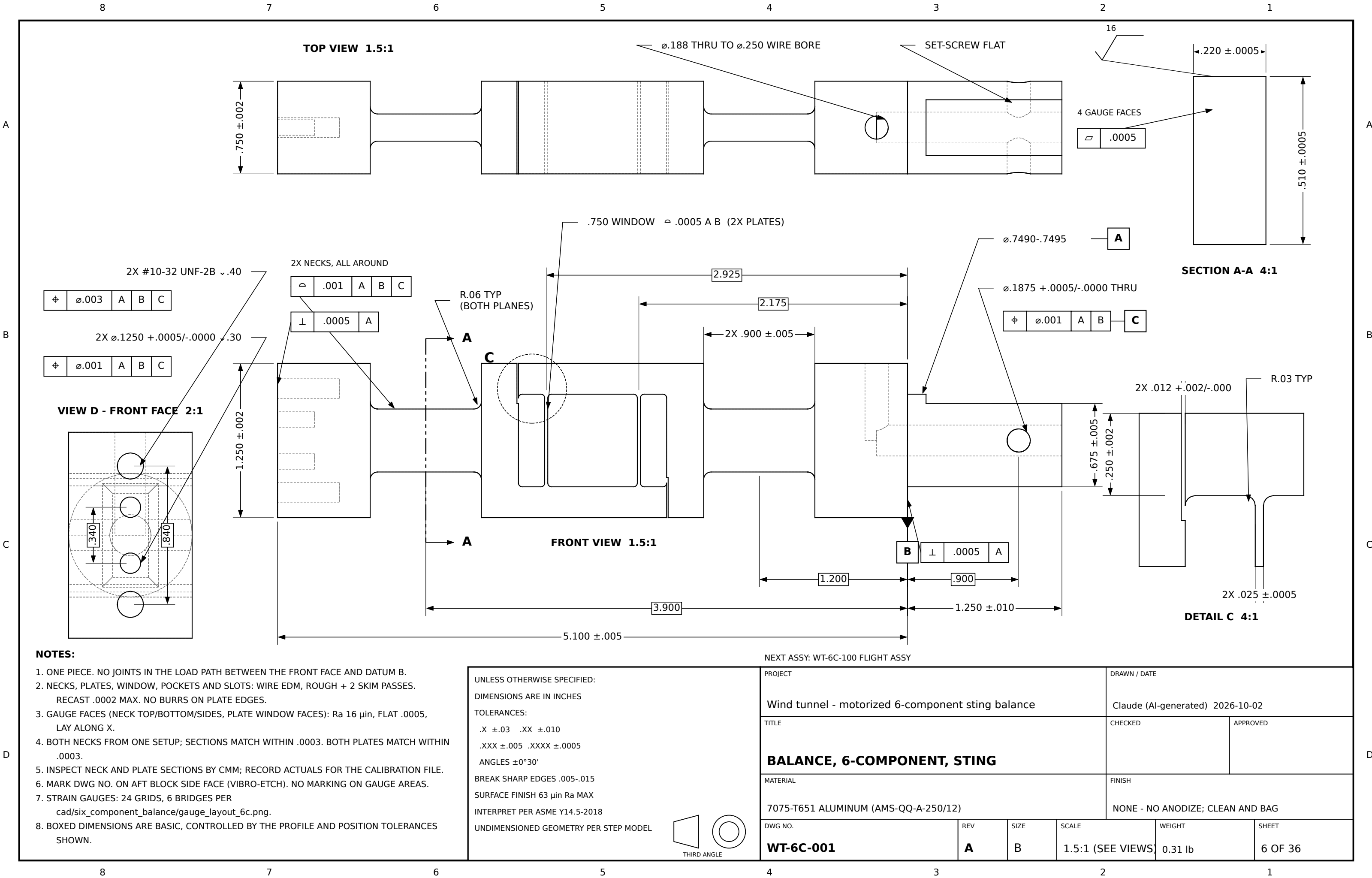


PRELIMINARY - TO CONFIRM

PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-501	Rail, yaw, inner	1	6061-T6
2	WT-6C-502	Rail, yaw, outer	1	6061-T6
3	WT-6C-503	Rack, yaw, floor	1	6061-T6
4	WT-6C-504	Plate, yaw carriage	1	6061-T6
5	WT-6C-505	Pinion, yaw (modified stock gear)	1	steel, bought
6	WT-6C-506-1	Spacer, wheel, plain	4	6061-T6
7	WT-6C-506-2	Spacer, wheel, eccentric	4	6061-T6
8	-	V-groove wheel 1.00 OD, 90 deg	8	bought
9	-	Wheel stud / shoulder bolt + nut	8	bought
10	-	Worm gearmotor 30:1 + NEMA 17	1	bought
11	WT-6C-400	Pitch sector assy	1	see WT-6C-400
12	-	FHCS 3/8 (rails, rack to floor)	26	bought
13	-	SHCS 3/8-16 (sector base to carriage)	4	bought
14	-	Home / limit switches	1	see Electronics

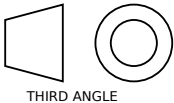
PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, YAW CARRIAGE								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.	REV	SIZE	SCALE	WEIGHT	SHEET			
WT-6C-500	A	B	NTS	-	5 OF 36			



NOTES:

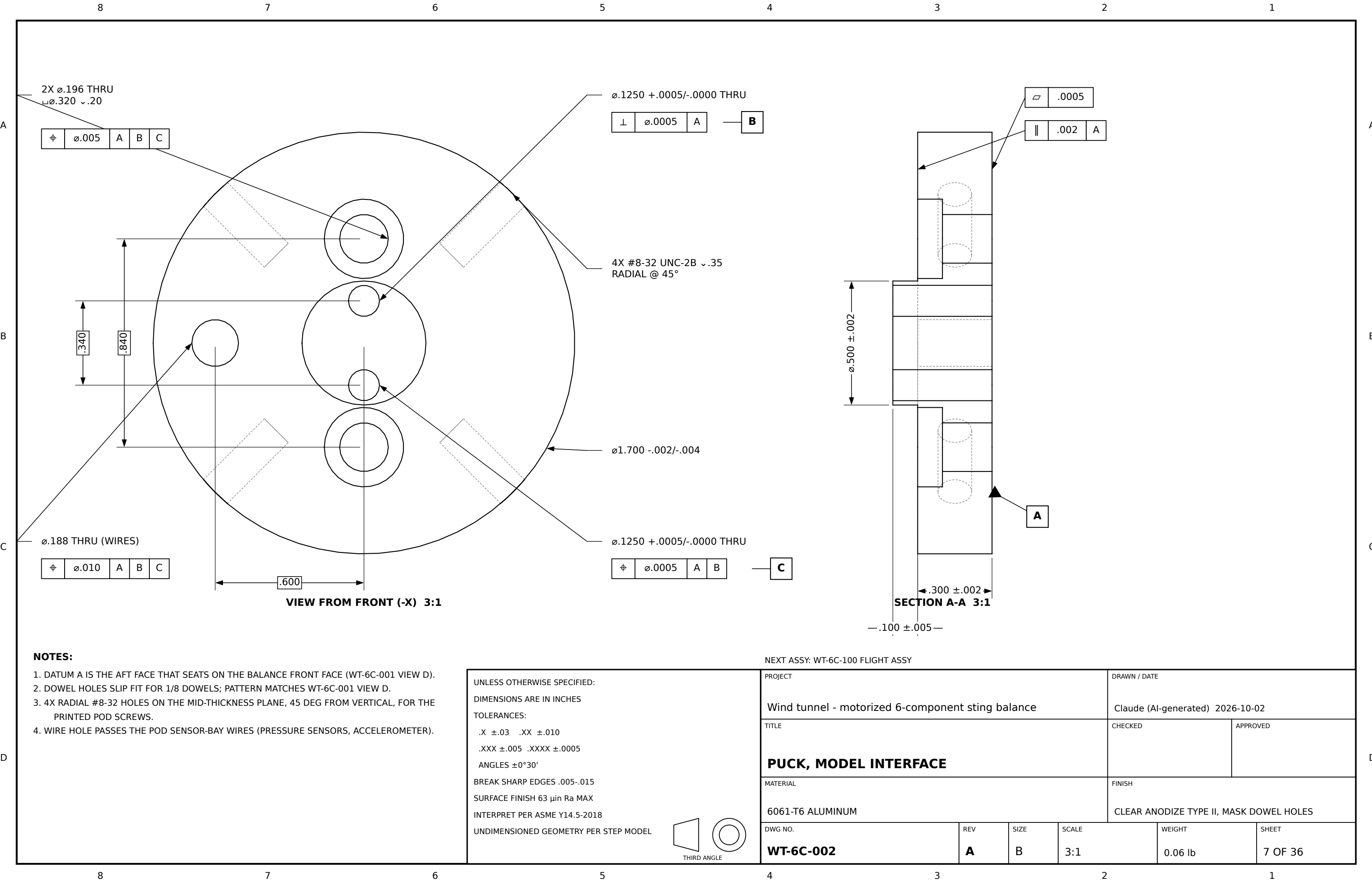
- ONE PIECE. NO JOINTS IN THE LOAD PATH BETWEEN THE FRONT FACE AND DATUM B.
- NECKS, PLATES, WINDOW, POCKETS AND SLOTS: WIRE EDM, ROUGH + 2 SKIM PASSES.
RECAST .0002 MAX. NO BURRS ON PLATE EDGES.
- GAUGE FACES (NECK TOP/BOTTOM/SIDES, PLATE WINDOW FACES): Ra 16 µin, FLAT .0005, LAY ALONG X.
- BOTH NECKS FROM ONE SETUP; SECTIONS MATCH WITHIN .0003. BOTH PLATES MATCH WITHIN .0003.
- INSPECT NECK AND PLATE SECTIONS BY CMM; RECORD ACTUALS FOR THE CALIBRATION FILE.
- MARK DWG NO. ON AFT BLOCK SIDE FACE (VIBRO-ETCH). NO MARKING ON GAUGE AREAS.
- STRAIN GAUGES: 24 GRIDS, 6 BRIDGES PER
cad/six_component_balance/gauge_layout_6c.png.
- BOXED DIMENSIONS ARE BASIC, CONTROLLED BY THE PROFILE AND POSITION TOLERANCES SHOWN.

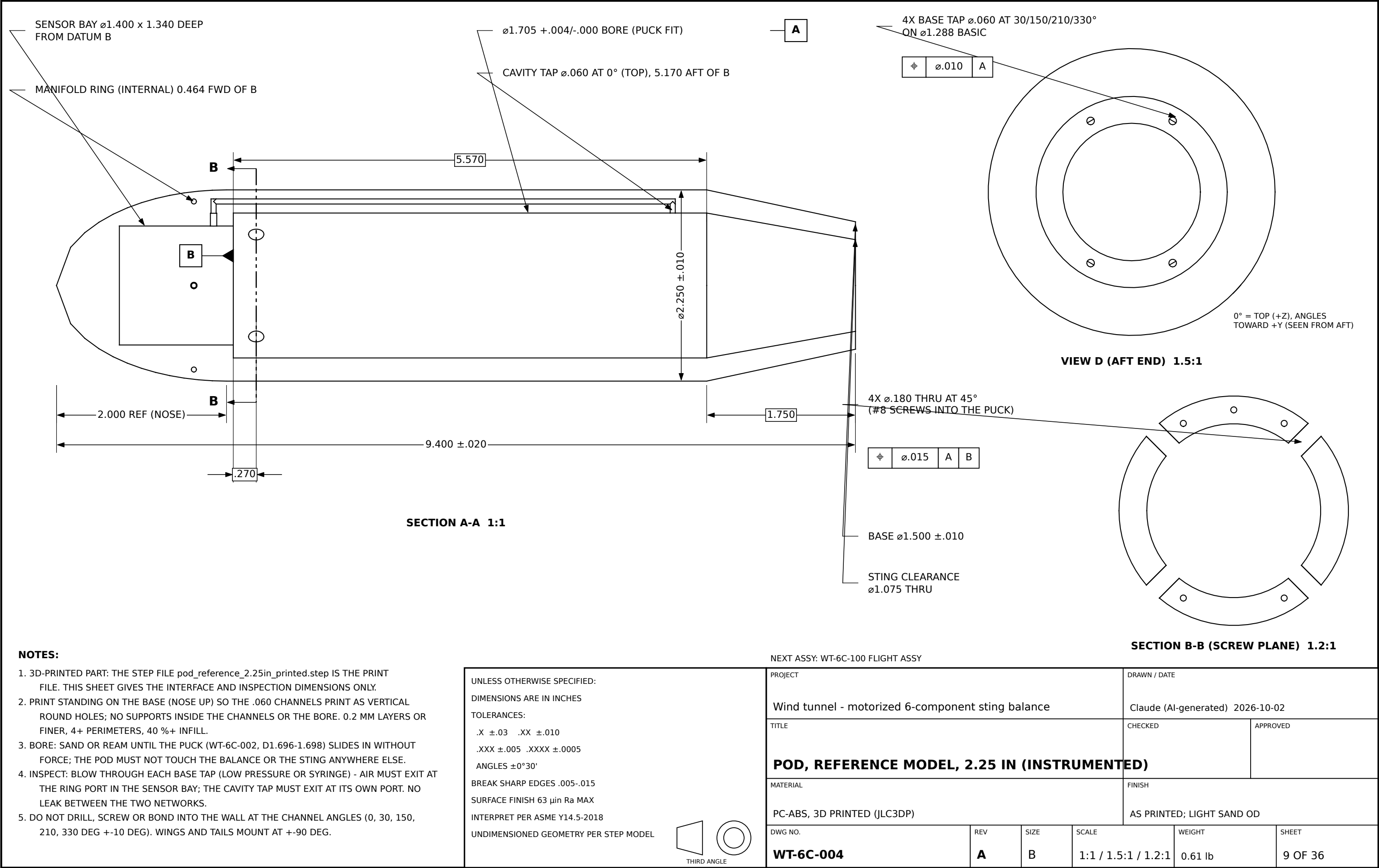
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X ±.03 .XX ±.010
.XXX ±.005 .XXXX ±.0005
ANGLES ±0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 µin Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL

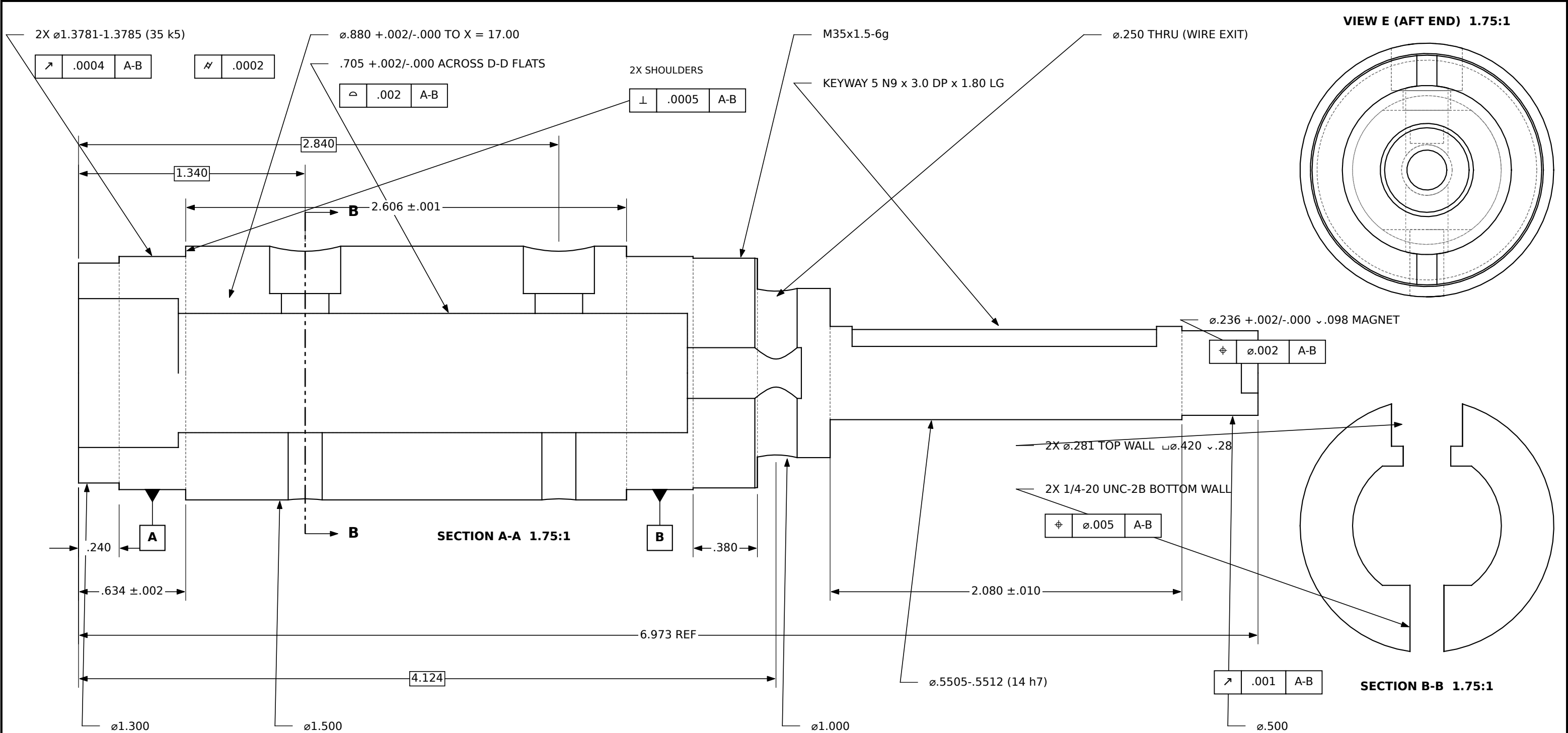


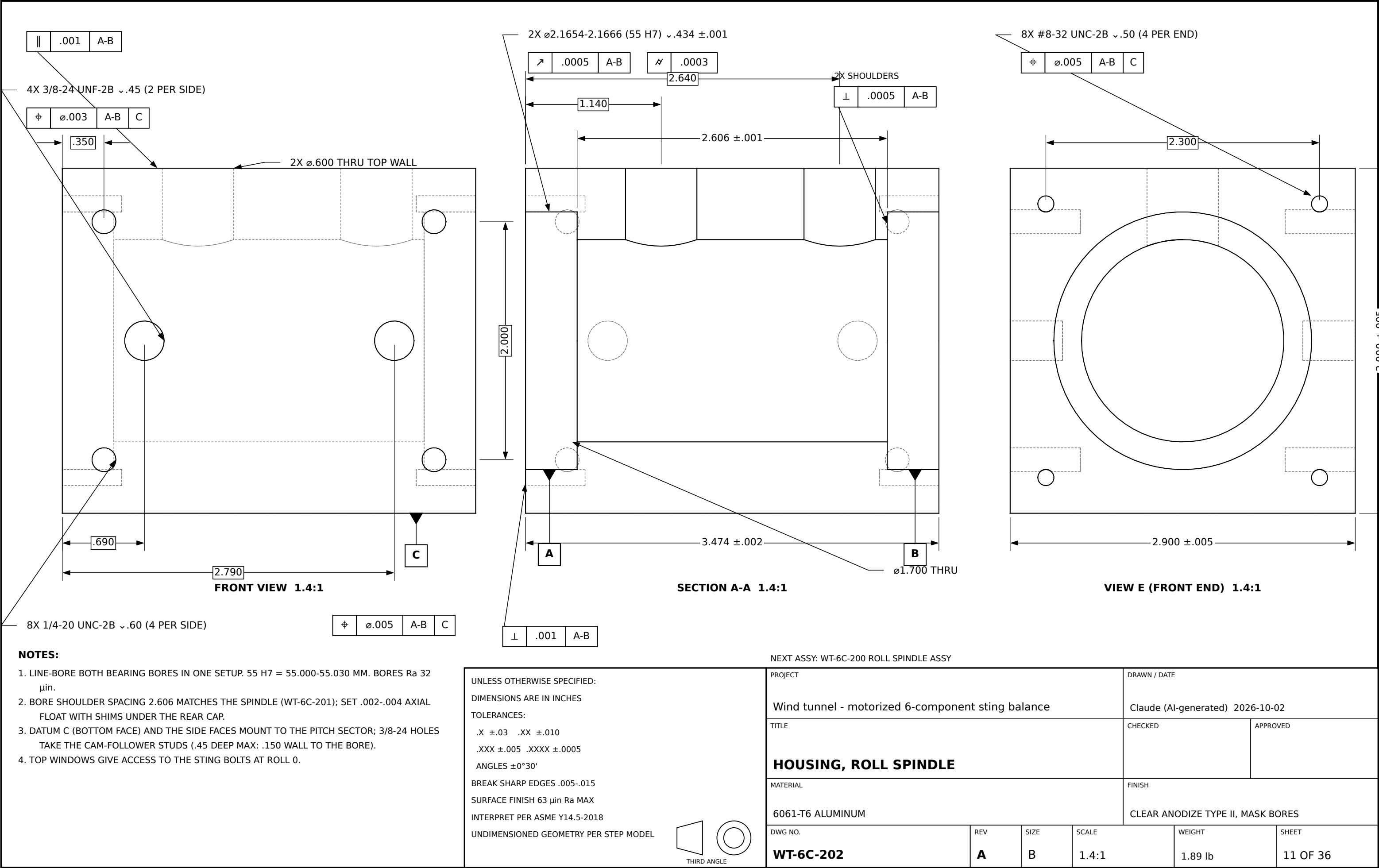
NEXT ASSY: WT-6C-100 FLIGHT ASSY

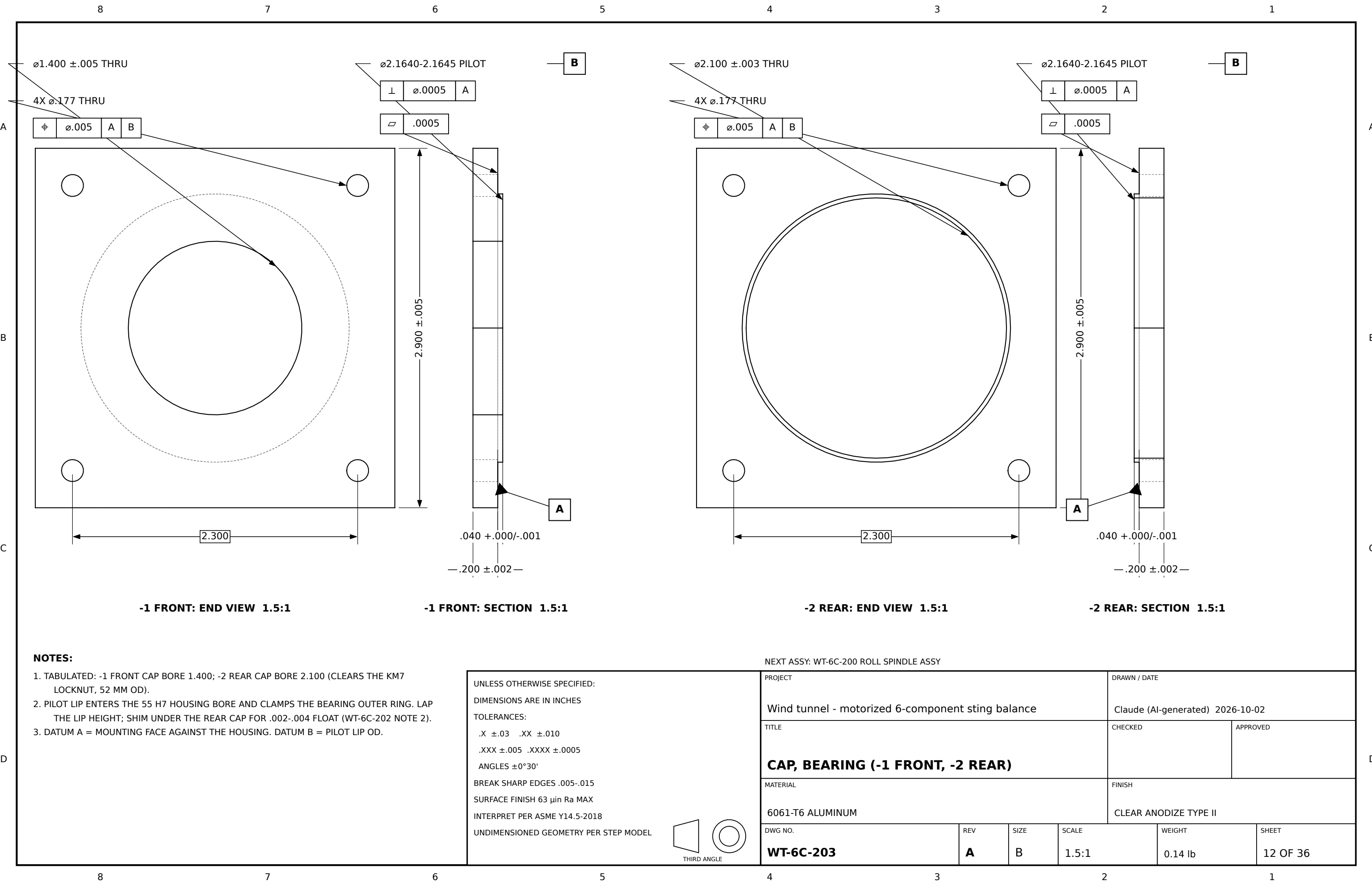
PROJECT				DRAWN / DATE		
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02		
TITLE				CHECKED		APPROVED
BALANCE, 6-COMPONENT, STING						
MATERIAL				FINISH		
7075-T651 ALUMINUM (AMS-QQ-A-250/12)				NONE - NO ANODIZE; CLEAN AND BAG		
DWG NO.	REV	SIZE	SCALE	WEIGHT		SHEET
WT-6C-001	A	B	1.5:1 (SEE VIEWS)	0.31 lb		6 OF 36

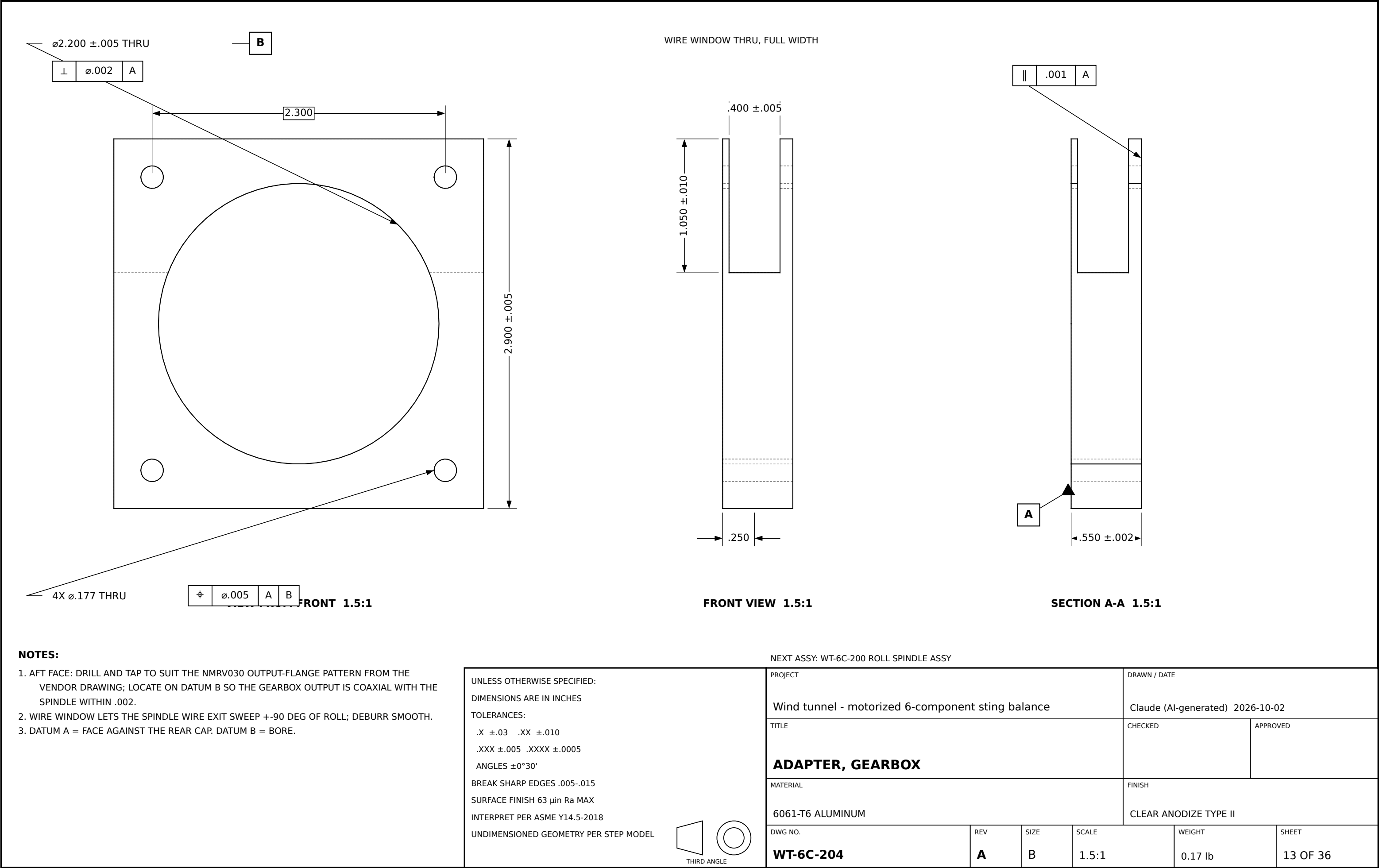


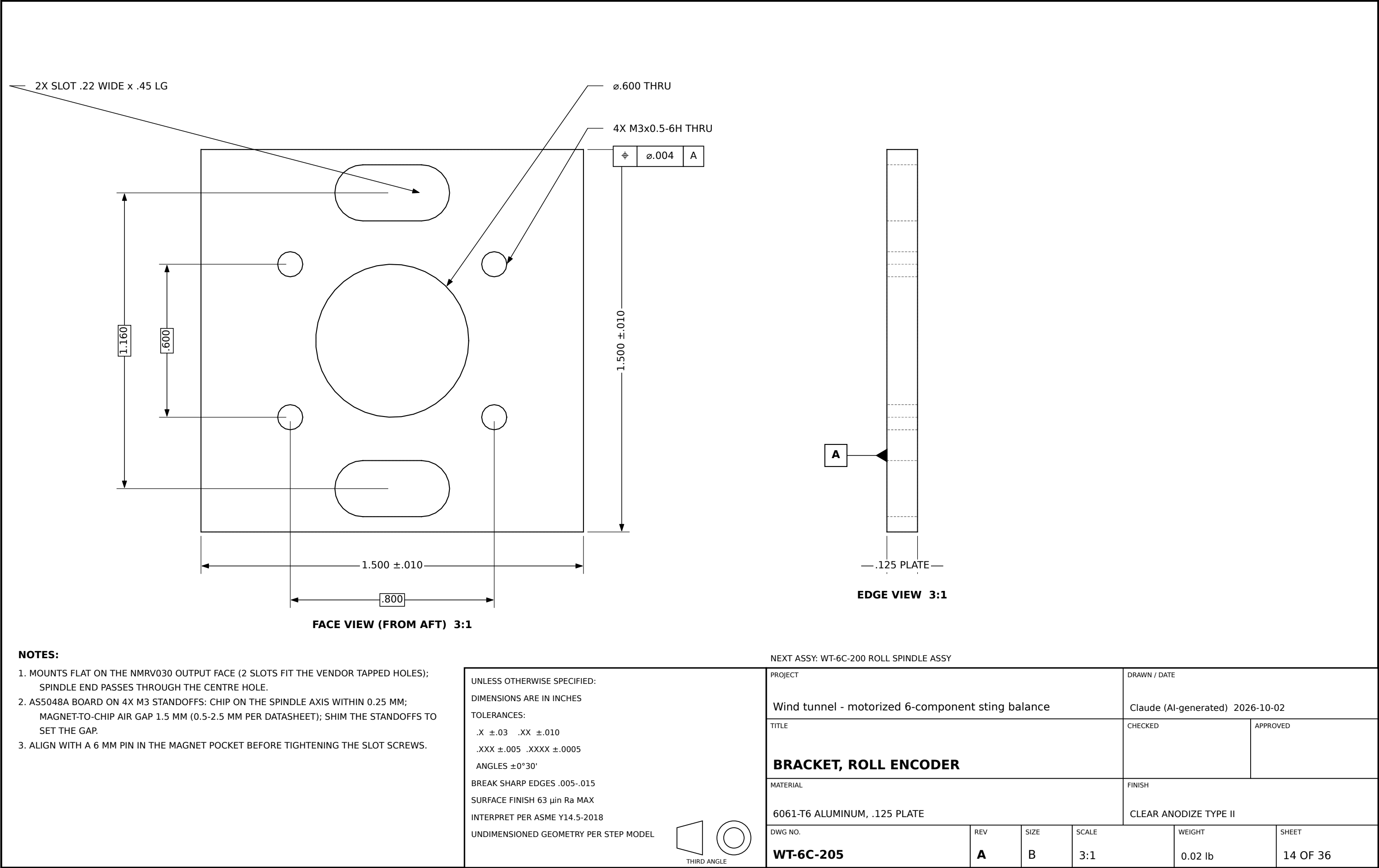


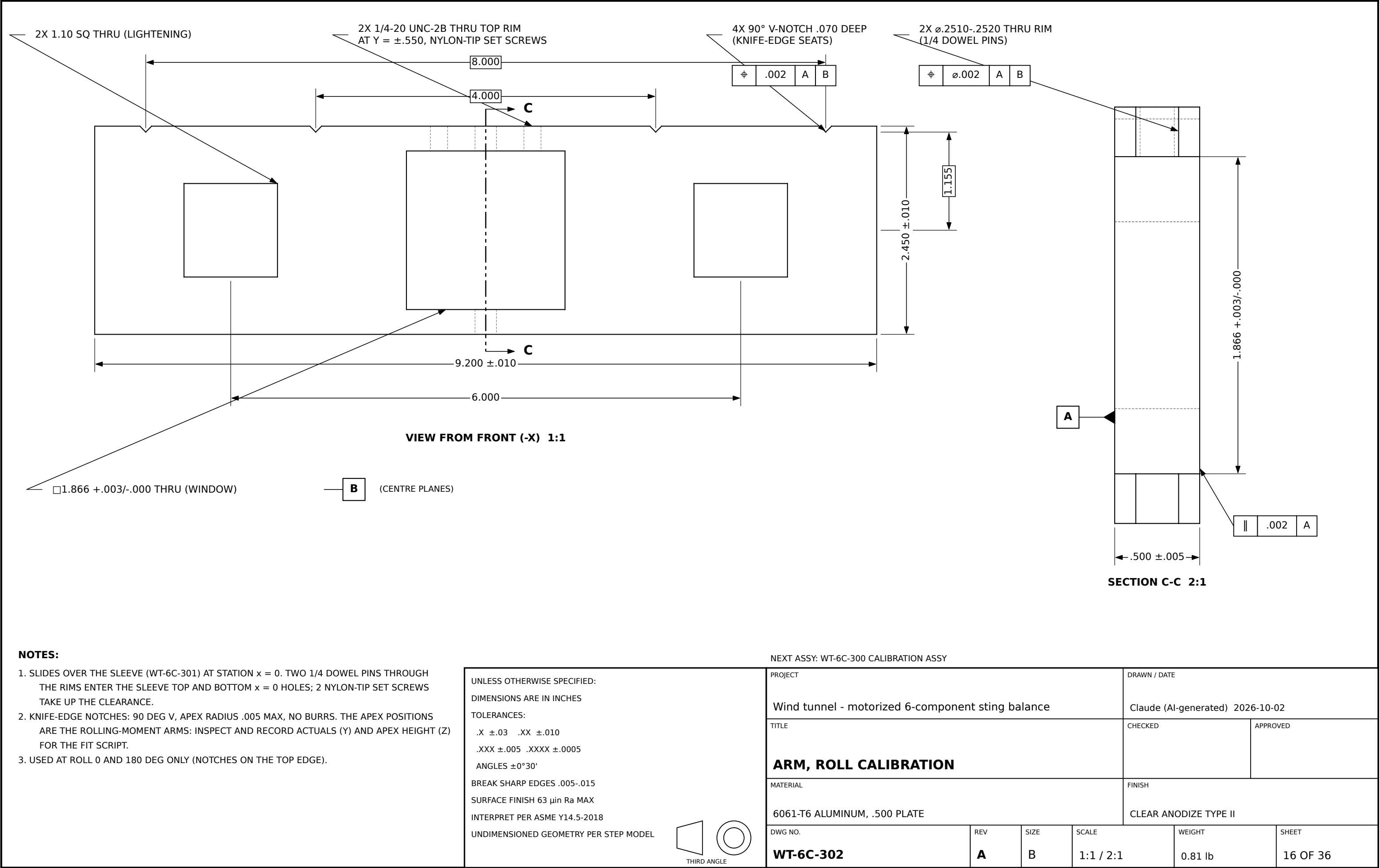


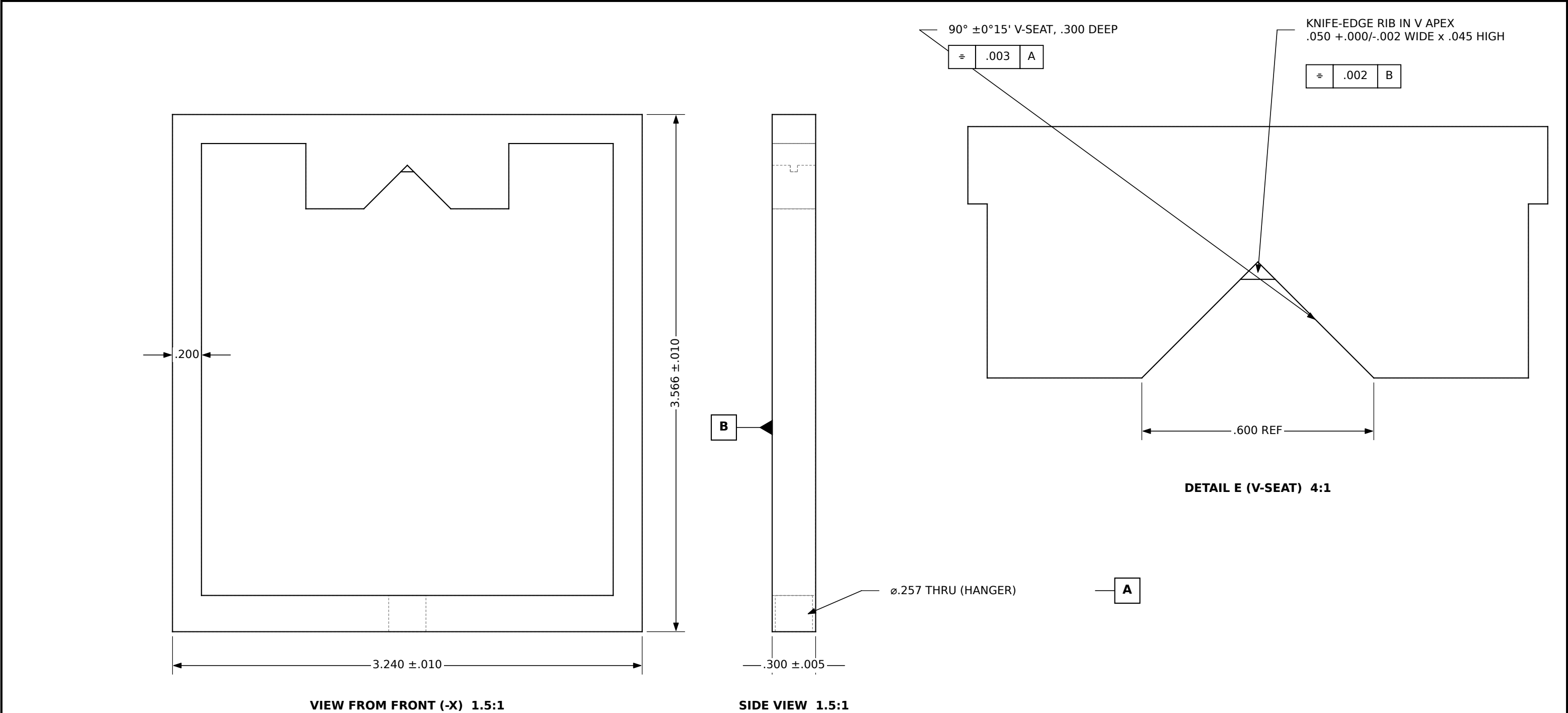












NOTES:

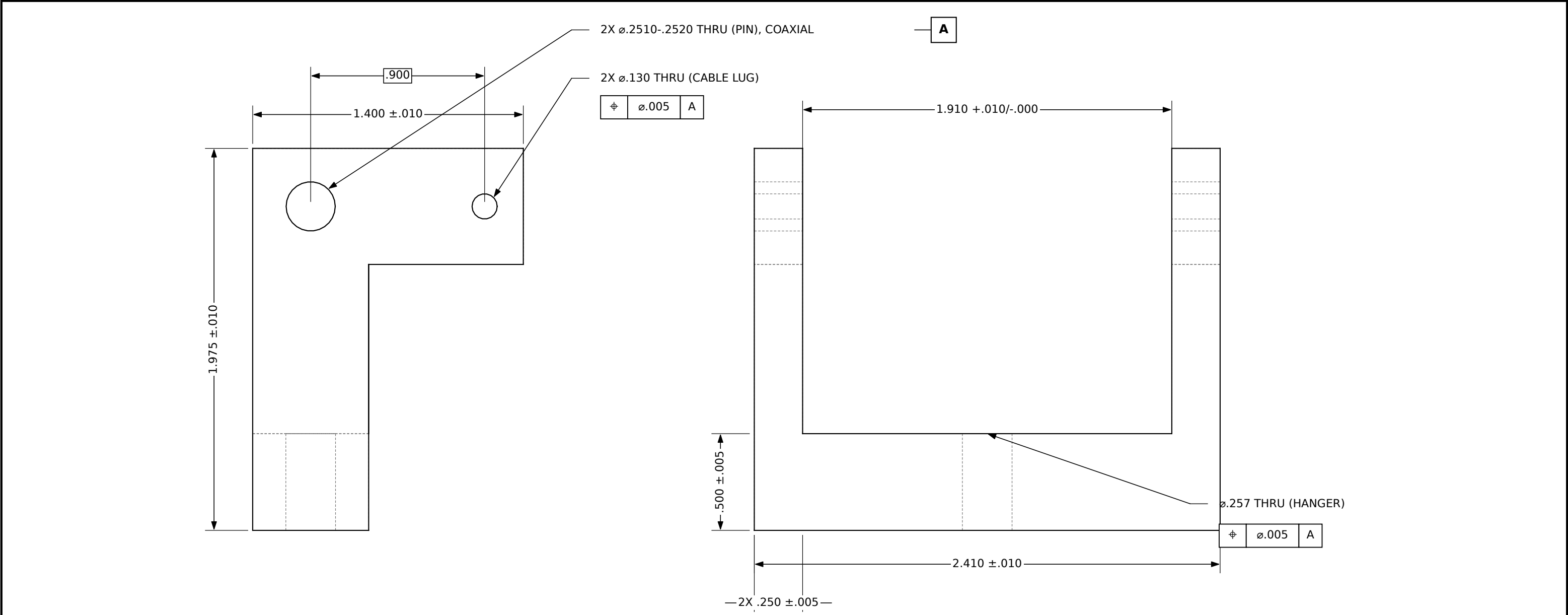
1. PROFILE: WATERJET OR MILL FROM .300 PLATE, THEN MILL THE V-SEAT AND THE KNIFE-EDGE RIB IN ONE SETUP.
2. SITS ON THE UPPER CORNER OF THE SLEEVE (WT-6C-301) AT ROLL 45 / 135 / 315 DEG; THE RIB DROPS INTO THE .060 CORNER GROOVE AT THE STATION. HANGER LOAD LINE PASSES THROUGH THE V APEX AND THE BALANCE AXIS (NO ROLLING MOMENT).
3. V-SEAT FACES Ra 32 μin, NO BURRS; RIB EDGES BROKEN .005 MAX.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X ±.03 .XX ±.010
.XXX ±.005 .XXXX ±.0005
ANGLES ±0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 μin Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL

THIRD ANGLE

NEXT ASSY: WT-6C-300 CALIBRATION ASSY

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
STIRRUP, 45 DEG CALIBRATION							
MATERIAL				FINISH			
6061-T6 ALUMINUM, .300 PLATE				CLEAR ANODIZE TYPE II			
DWG NO.		REV	SIZE	SCALE		WEIGHT	SHEET
WT-6C-303		A	B	1.5:1 / 4:1		0.09 lb	17 OF 36



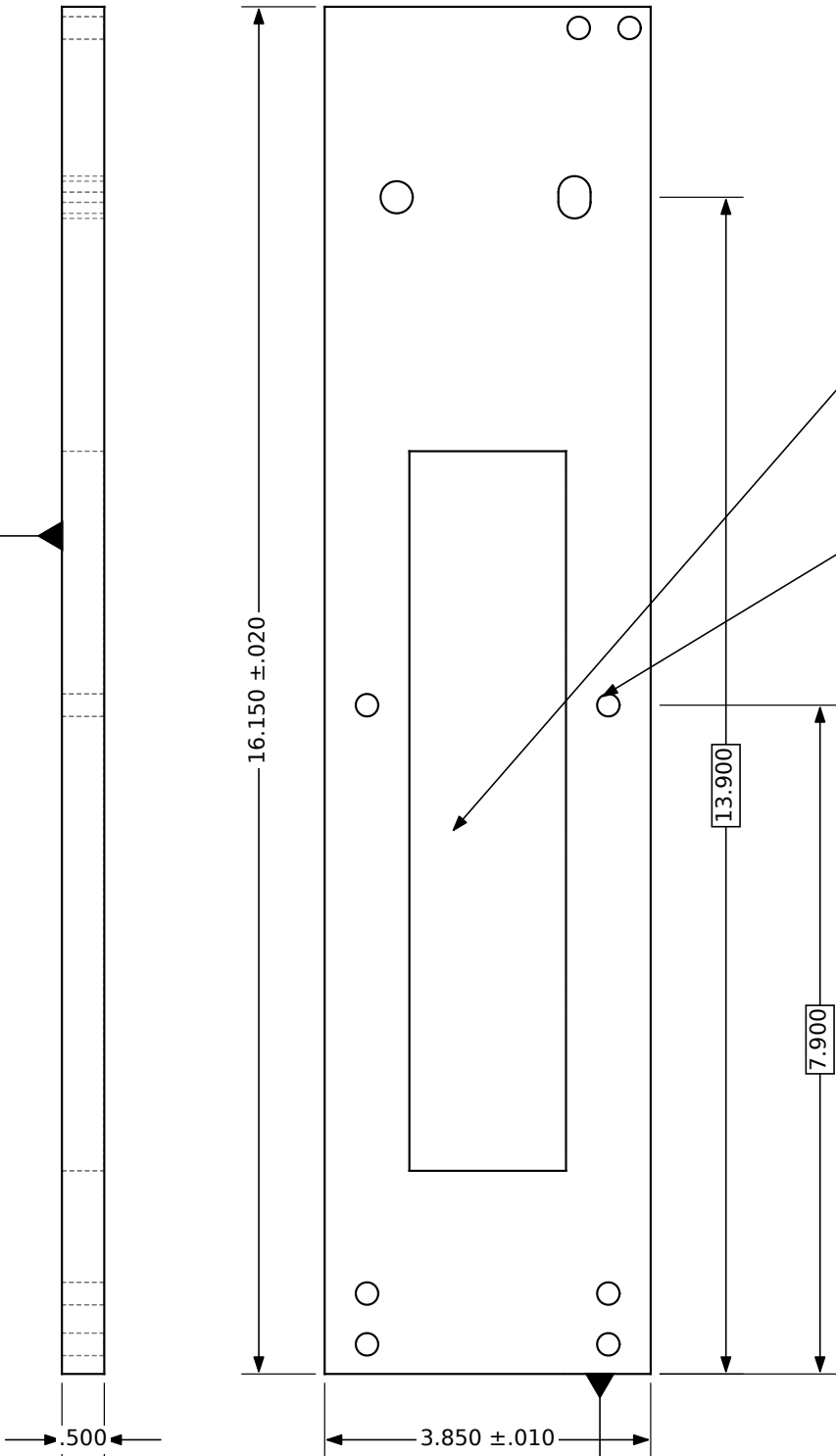
FRONT VIEW 2:1

VIEW FROM FRONT (-X) 2:1

- NOTES:**
- 1. MAY BE MADE AS 3 PIECES (2 STRAPS + BAR) JOINED WITH 2X #8-32 SHCS + 1/8 DOWEL PER JOINT; LINE-REAM THE PIN HOLES AFTER ASSEMBLY.
 - 2. HANGS ON A 1/4 GROUND PIN THROUGH THE SLEEVE STATION HOLES; ITS WEIGHT ACTS AT THE PIN AXIS. WEIGH THE FINISHED YOKE AND UPDATE analysis/calibration_6c_mass_properties.json.
 - 3. SAME YOKE FOR ROLL 0 / 90 / 180 / 270 DEG (ON WHICHEVER PAIR OF SLEEVE WALLS IS HORIZONTAL).
 - 4. CABLE LUGS TAKE THE DRAG CABLE (LAB +X) AT THE SAME STATION.

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: .X ±.03 .XX ±.010 .XXX ±.005 .XXXX ±.0005 ANGLES ±0°30' BREAK SHARP EDGES .005-.015 SURFACE FINISH 63 µin Ra MAX INTERPRET PER ASME Y14.5-2018 UNDIMENSIONED GEOMETRY PER STEP MODEL THIRD ANGLE	PROJECT				DRAWN / DATE			
	Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
	TITLE				CHECKED		APPROVED	
	YOKE, CALIBRATION, 6-COMPONENT							
	MATERIAL				FINISH			
6061-T6 ALUMINUM				CLEAR ANODIZE TYPE II				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-304		A	B	2:1		0.13 lb		18 OF 36

- NOTES:**
1. QTY 2, IDENTICAL (ONE EACH SIDE OF THE ROLL HOUSING).
 2. PIVOT HOLE AND LEVELING SLOT TAKE PLAIN 3/8 STUDS IN THE ROLL HOUSING CAM-FOLLOWER HOLES (WT-6C-202); THE SLOT GIVES +/-1.6 DEG OF LEVELING.
 3. DATUM A = INNER FACE: BOLTS AGAINST THE BASE TUBE SIDE WALL (WT-6C-307), .050 GAP TO THE HOUSING.
 4. MACHINE THE PIVOT HOLE AND SLOT WITH BOTH PLATES STACKED AND PINNED (ONE SETUP).



SIDE VIEW + FRONT VIEW 1:2.2

WINDOW 1.85 x 8.50 THRU
(LIGHTENING)

2X $\varnothing$.266 THRU
(CROSS SPACERS)

$\varnothing$	.010	A	B	C
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UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X $\pm$.03 .XX $\pm$.010

.XXX $\pm$.005 .XXXX $\pm$.0005

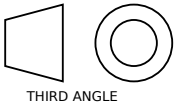
ANGLES $\pm$ 0°30'

BREAK SHARP EDGES .005-.015

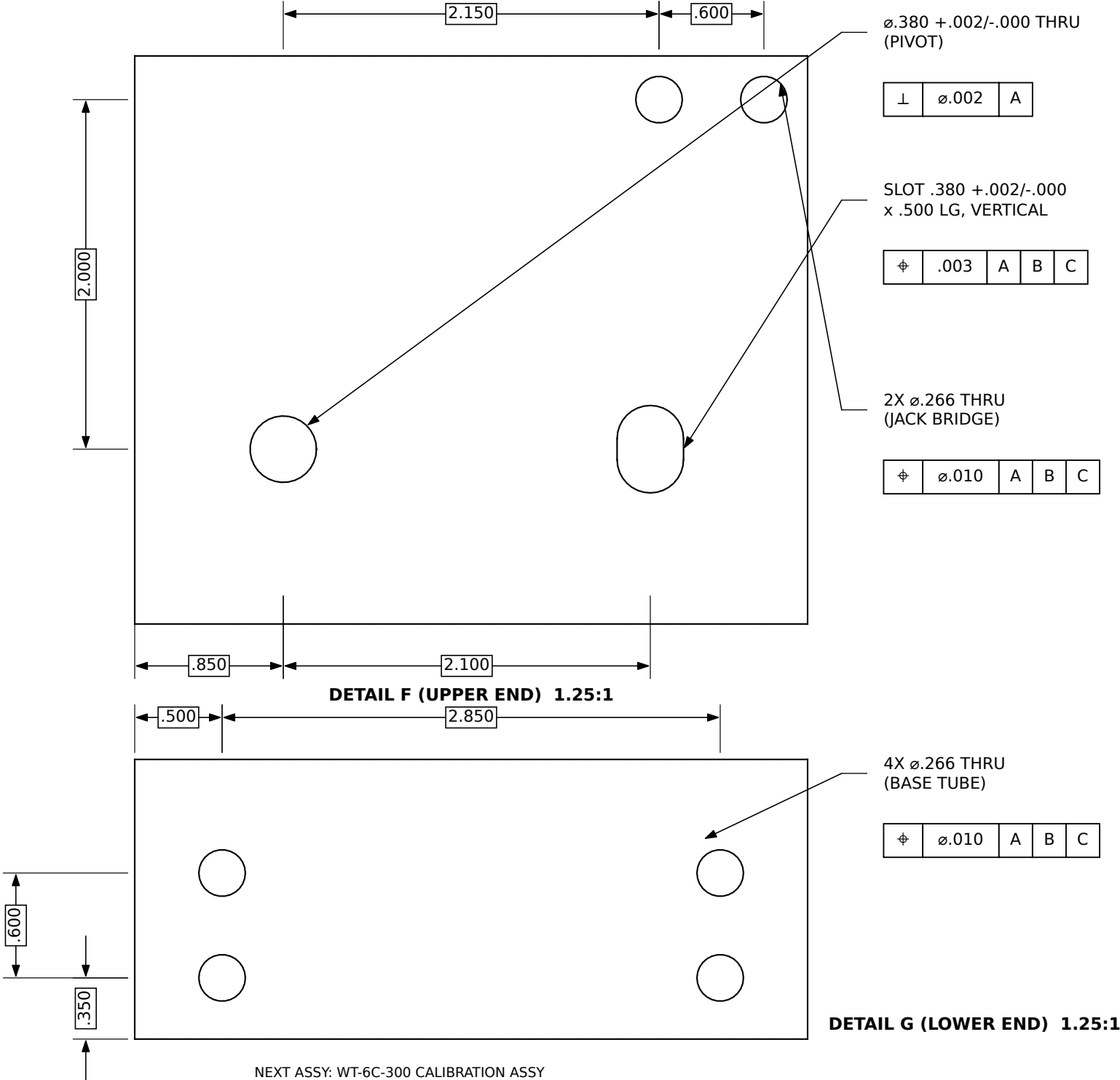
SURFACE FINISH 63 μ in Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



THIRD ANGLE



$\varnothing$.380 $\pm$.002/- .000 THRU
(PIVOT)

$\perp$	$\varnothing$.002	A
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SLOT .380 $\pm$.002/- .000
x .500 LG, VERTICAL

$\varnothing$	.003	A	B	C
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2X $\varnothing$.266 THRU
(JACK BRIDGE)

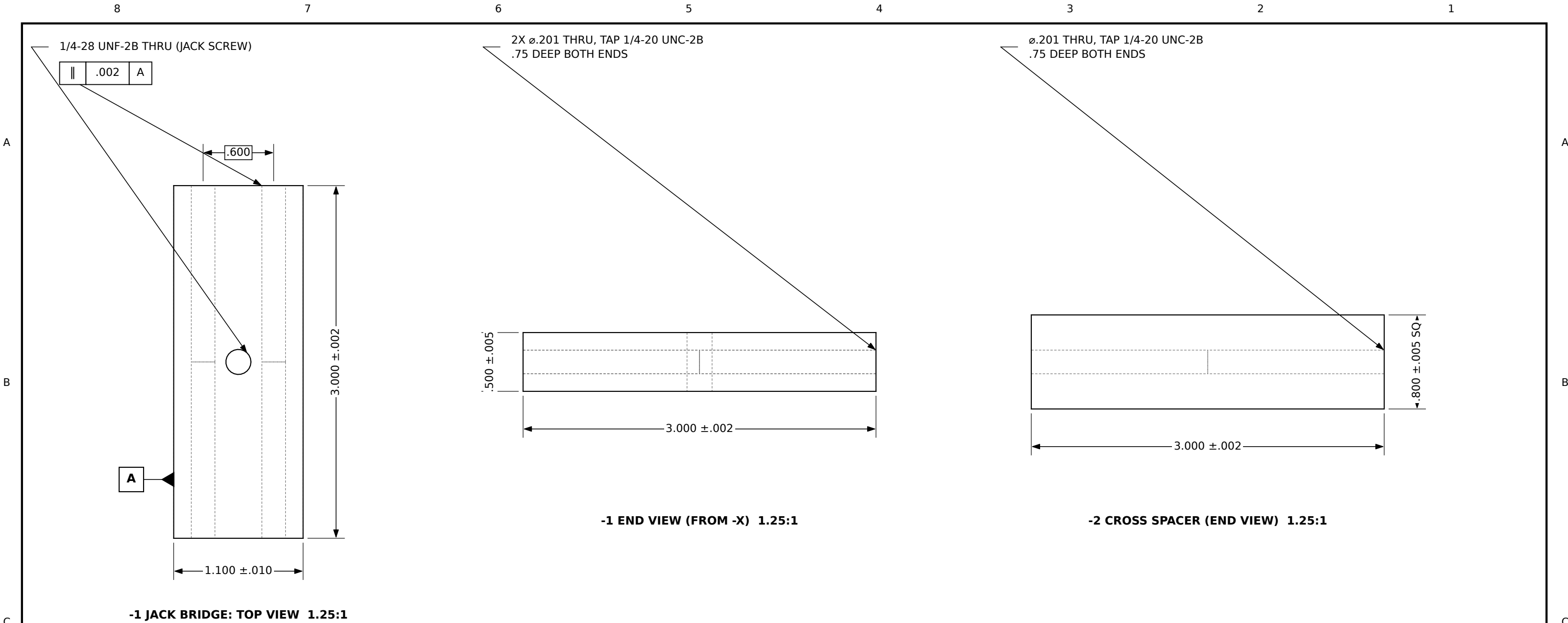
$\varnothing$	.010	A	B	C
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4X $\varnothing$.266 THRU
(BASE TUBE)

$\varnothing$	.010	A	B	C
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NEXT ASSY: WT-6C-300 CALIBRATION ASSY

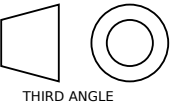
PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
PLATE, HEAD, CALIBRATION STAND (QTY 2)								
MATERIAL				FINISH				
6061-T6 ALUMINUM, .500 PLATE				CLEAR ANODIZE TYPE II				
DWG NO.	REV	SIZE	SCALE		WEIGHT			
WT-6C-305	A	B	1:2.2 / 1.25:1		2.23 lb			
						SHEET		
						19 OF 36		



NOTES:

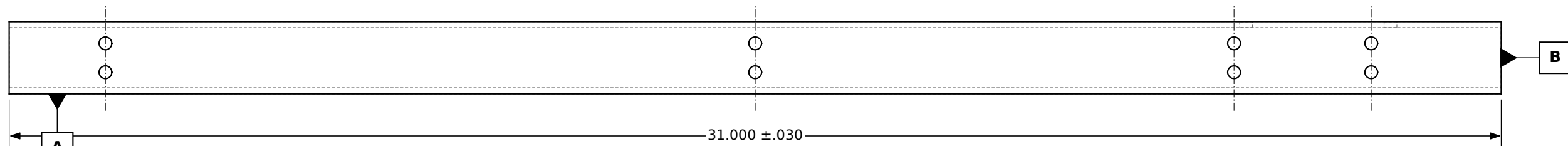
1. -1 JACK BRIDGE SPANS THE HEAD PLATES (WT-6C-305) ABOVE THE ROLL HOUSING. THE 1/4-28 JACK SCREW (BRASS TIP) BEARS ON THE HOUSING TOP: .036 IN TRAVEL PER TURN = .53 DEG OF LEVEL PER TURN.
2. -2 CROSS SPACERS SET THE HEAD-PLATE GAP (3.000) AT MID-HEIGHT. QTY 2.
3. END FACES OF BOTH PARTS PARALLEL .002: THEY SET THE PLATE SPACING.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X $\pm$.03 .XX $\pm$.010
.XXX $\pm$.005 .XXXX $\pm$.0005
ANGLES $\pm$ 0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 μ in Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL



PROJECT					DRAWN / DATE		
Wind tunnel - motorized 6-component sting balance					Claude (AI-generated) 2026-10-02		
TITLE					CHECKED		APPROVED
BRIDGE, JACK (-1) / SPACER, CROSS (-2, QTY 2)							
MATERIAL					FINISH		
6061-T6 ALUMINUM					CLEAR ANODIZE TYPE II		
DWG NO.	REV	SIZE	SCALE	WEIGHT	SHEET		
WT-6C-306	A	B	1.25:1	0.14 lb	20 OF 36		

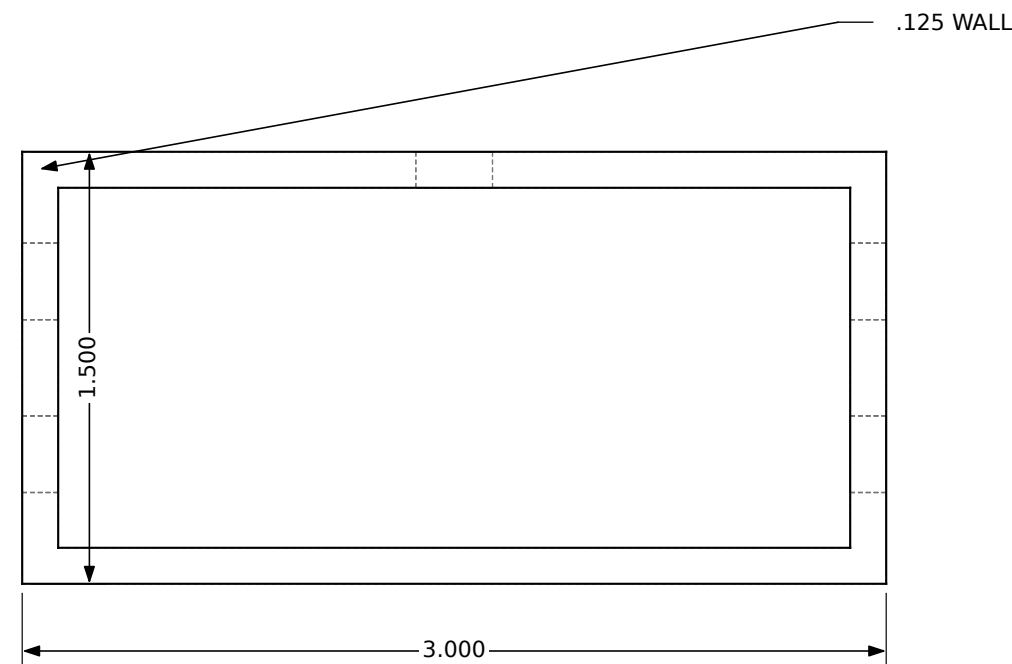
A



FRONT VIEW 1:2.5

HOLE TABLE - ALL Ø.266 THRU; POSITION Ø.010 A B (BASIC DIMS)

HOLES	WALL	FROM B (X)	FROM A (Z)	PURPOSE
H1-H4	SIDE WALLS THRU	29.000	.450 / 1.050	THRUST UPRIGHT
H5-H8	SIDE WALLS THRU	15.500	.450 / 1.050	DRAG UPRIGHT
H9-H12	SIDE WALLS THRU	5.550	.450 / 1.050	HEAD PLATE (NEW)
H13-H16	SIDE WALLS THRU	2.700	.450 / 1.050	HEAD PLATE (NEW)
H17	TOP WALL	5.300	-	3-COMP FOOT BLOCK (OPTIONAL)
H18	TOP WALL	2.300	-	3-COMP FOOT BLOCK (OPTIONAL)



END VIEW 1.5:1

NOTES:

1. CUT TO LENGTH, ENDS SQUARE .005. ALL HOLES Ø.266 (1/4 BOLTS). SIDE-WALL HOLES DRILLED THROUGH BOTH WALLS IN ONE PASS.
2. HOLE TABLE DIMENSIONS FROM THE AFT END (DATUM B) AND THE BOTTOM FACE (DATUM A).
3. PULLEY UPRIGHT HOLES (DRAG $x = 7.0$, THRUST $x = -6.5$ FROM THE BMC) ARE UNCHANGED FROM THE 3-COMPONENT STAND; HEAD-PLATE HOLES ARE NEW FOR THE 6-COMPONENT HEAD.
4. CLAMP TO THE BENCH WITH 4 C-CLAMPS OR BOLT THROUGH THE BOTTOM WALL; LEVEL THE TOP WITHIN .02 DEG.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X $\pm$.03 .XX $\pm$.010

.XXX ±.005 .XXXX ±.0005

ANGLES $\pm 0^{\circ}30'$

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 μin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



THIRD ANGLE

NEXT ASSY: WT-6C-300 CALIBRATION ASSY

PROJECT				DRAWN / DATE		
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02		
TITLE				CHECKED		APPROVED
MATERIAL				FINISH		
6061-T6 RECT. TUBE 3.00 x 1.50 x .125 WALL				AS SUPPLIED; DEBURR		
DWG NO.	REV	SIZE	SCALE	WEIGHT	SHEET	
WT-6C-307	A	B	1:2.5 / 1.5:1	3.20 lb	21 OF 36	

8

7

6

5

4

3

2

1

D

□

C

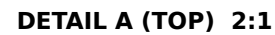
E

A

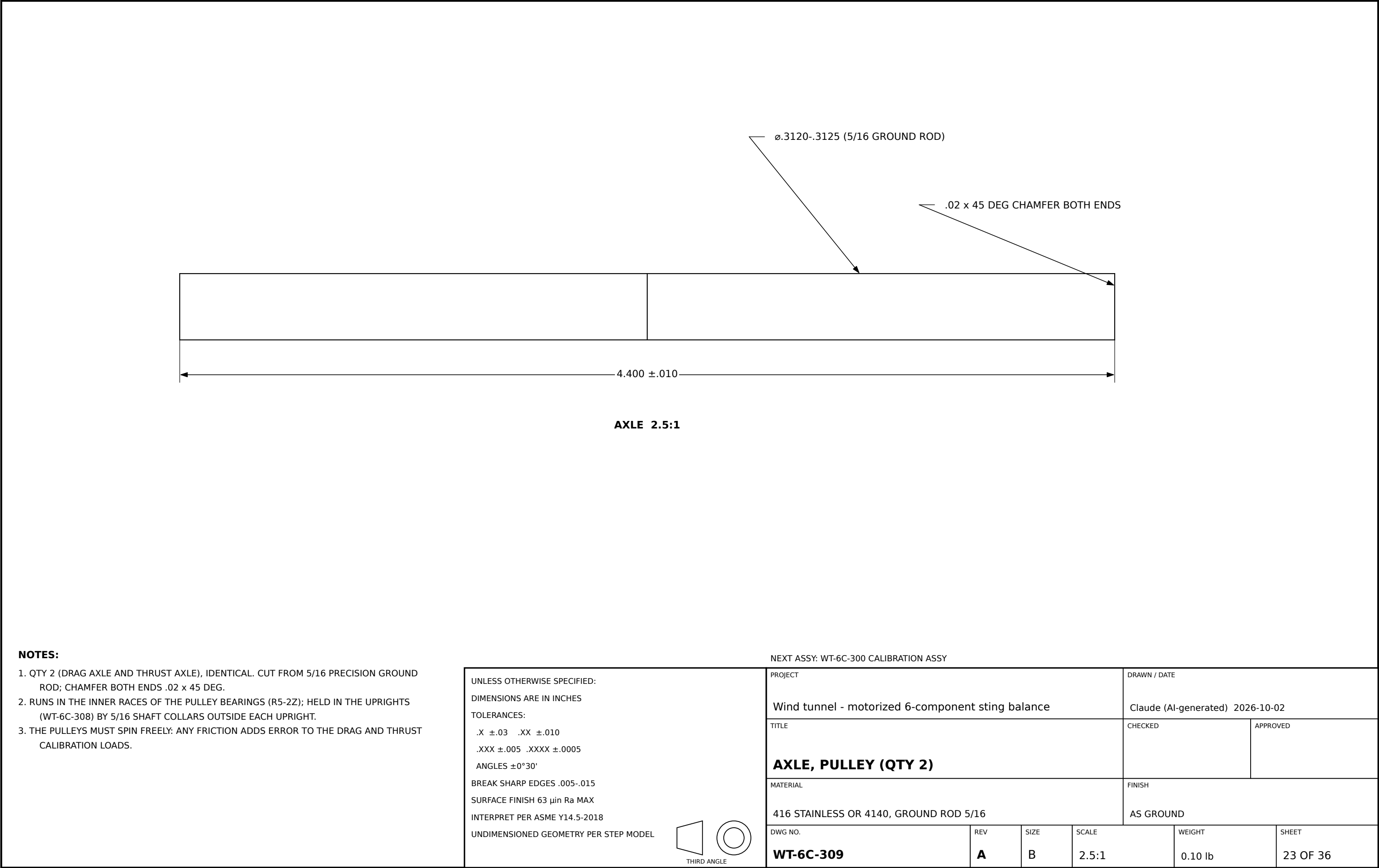
D



1. QTY 4, IDENTICAL: 2 FOR THE DRAG AXLE (X = +7.0 FROM THE BMC), 2 FOR THE THRUST AXLE (X = -6.5).
2. BOLTS TO THE BASE-TUBE SIDE WALLS (WT-6C-307) WITH 2X 1/4-20 SHCS + NUTS.
3. THE AXLE HOLE HEIGHT SETS THE CABLE LINE ON THE BALANCE AXIS: KEEP 12.950 BASIC FROM THE BOTTOM EDGE.
4. MACHINE ALL FOUR STACKED SO THE AXLE HOLES LINE UP.



PROJECT					DRAWN / DATE		
Wind tunnel - motorized 6-component sting balance					Claude (AI-generated) 2026-10-02		
TITLE					CHECKED		APPROVED
UPRIGHT, PULLEY (QTY 4)							
MATERIAL					FINISH		
6061-T6 ALUMINUM, .375 PLATE					CLEAR ANODIZE TYPE II		
DWG NO.			REV	SIZE	SCALE	WEIGHT	SHEET
WT-6C-308			A	B	1:2 / 2:1	0.74 lb	22 OF 36



HOLE TABLE (BASIC, FROM THE PITCH AXIS) POSITION $\varnothing.005$ A B C

HOLE	SIZE	X	Z
H1	ø.281 THRU (TIE)	17.353	4.391
H2	ø.281 THRU (TIE)	18.710	4.734
C	ø.281 THRU (SPACER)	14.600	-8.600
H4	ø.281 THRU (SPACER)	17.000	-8.650
H5	1/4-20 x .75 DP (EDGE)	14.200	-9.100
H6	1/4-20 x .75 DP (EDGE)	15.600	-9.100
H7	1/4-20 x .75 DP (EDGE)	17.400	-9.100

$\varnothing$	$\varnothing.005$	A	B	C
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2X ARC SLOT .752 $+.002/-.000$ WIDE
FULL R ENDS

$\frac{1}{2}$	.002	A	B	C
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PITCH AXIS (BMC)
CENTRE OF ALL ARCS

FRONT VIEW 1:2.2

SIDE

NOTES:

1. QTY 2, IDENTICAL (+Y AND -Y). PROFILE AND SLOTS: CNC MILL (OR WATERJET + FINISH MILL THE SLOTS).
2. THE TWO ARC SLOTS ARE THE CAM-FOLLOWER TRACKS: THEIR CENTRE IS THE PITCH AXIS (BMC). MACHINE BOTH CHEEKS STACKED AND PINNED IN ONE SETUP. SLOT FACES Ra 32 μ m, NO STEPS AT TOOL ENTRY.
3. PITCH-AXIS ORIGIN: 9.100 ABOVE DATUM B AND 14.600 FORWARD OF HOLE C (BASIC). HOLE TABLE COORDINATES ARE FROM THE PITCH AXIS, X AFT, Z UP.
4. BOTTOM EDGE: 3X 1/4-20 UNC-2B .75 DEEP FOR THE BASE PLATE BOLTS (WT-6C-403).

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXX ±.0005

ANGLES $\pm 0^{\circ}30'$

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 μin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL

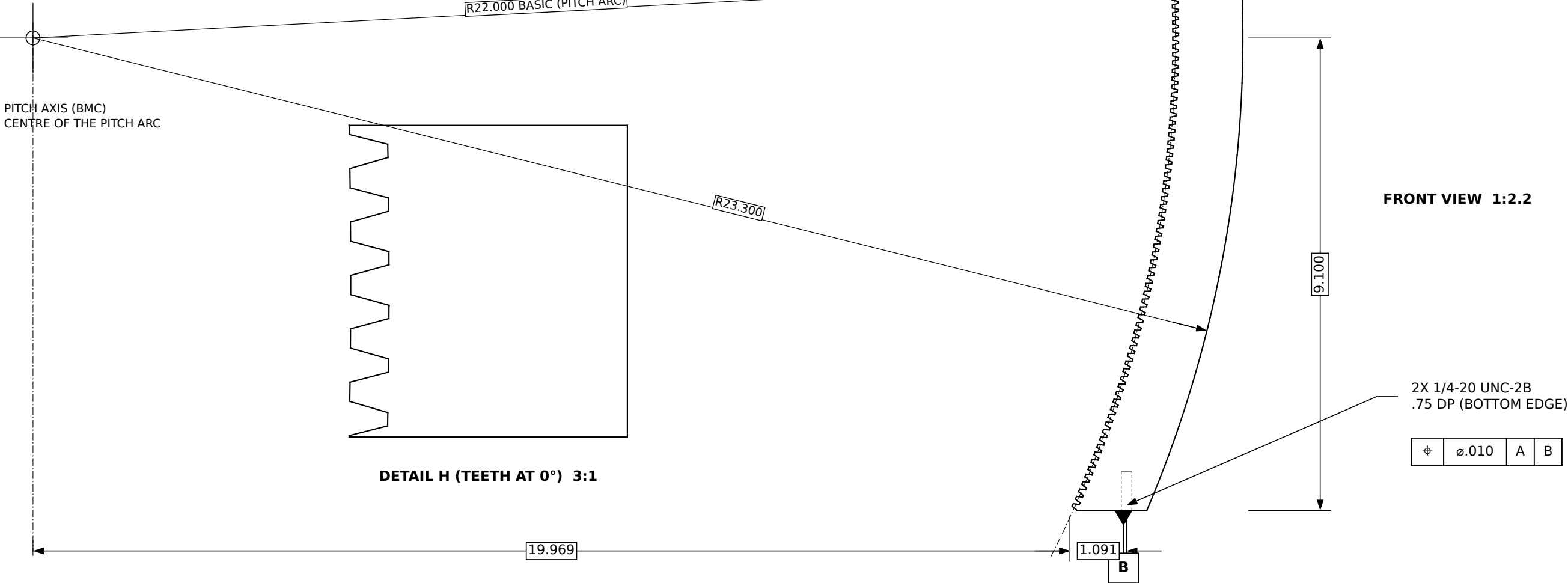


NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT					DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance					Claude (AI-generated) 2026-10-02			
TITLE					CHECKED		APPROVED	
CHEEK, PITCH SECTOR (QTY 2)								
MATERIAL					FINISH			
6061-T6 ALUMINUM, .500 PLATE					CLEAR ANODIZE TYPE II			
DWG NO.			REV	SIZE	SCALE	WEIGHT	SHEET	
WT-6C-401			A	B	1:2.2	2.59 lb	25 OF 36	

GEAR DATA

TYPE	INTERNAL ARC RACK (SEGMENT)
MODULE	1.25 mm
PRESSURE ANGLE	20°
PITCH RADIUS	22.000 in (558.8 mm) BASIC
TEETH ON SEGMENT	~87: FROM THE BASE TOP (-24.4°) TO +11°
ANGULAR PITCH	0.4026° BASIC
TOOTH THK AT PITCH	1.96 mm (-0.05/-0.10)
ADDENDUM / DEDENDUM	1.25 / 1.56 mm
MATING PINION	WT-6C-406, 20 T
QUALITY	AGMA Q7 / ISO 9



NOTES:

- TEETH: WIRE EDM (PREFERRED) OR CNC MILL WITH A MODULE 1.25 FORM CUTTER; PROFILE AND PROFILE FROM THE GEAR DATA TABLE. DEBURR TOOTH ENDS.
- THE PITCH ARC IS CENTRED ON THE PITCH AXIS (BMC), SAME ORIGIN AS THE CHEEKS (WT-6C-401): 9.100 ABOVE DATUM B.
- MATES THE 20-TOOTH PINION WT-6C-406. GRAVITY ALWAYS LOADS THE SAME TOOTH FLANK, SO BACKLASH DOES NOT REVERSE.
- FOOT: 2X 1/4-20 UNC-2B .75 DEEP INTO THE BOTTOM EDGE FOR THE BASE PLATE (WT-6C-403).

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

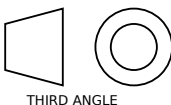
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

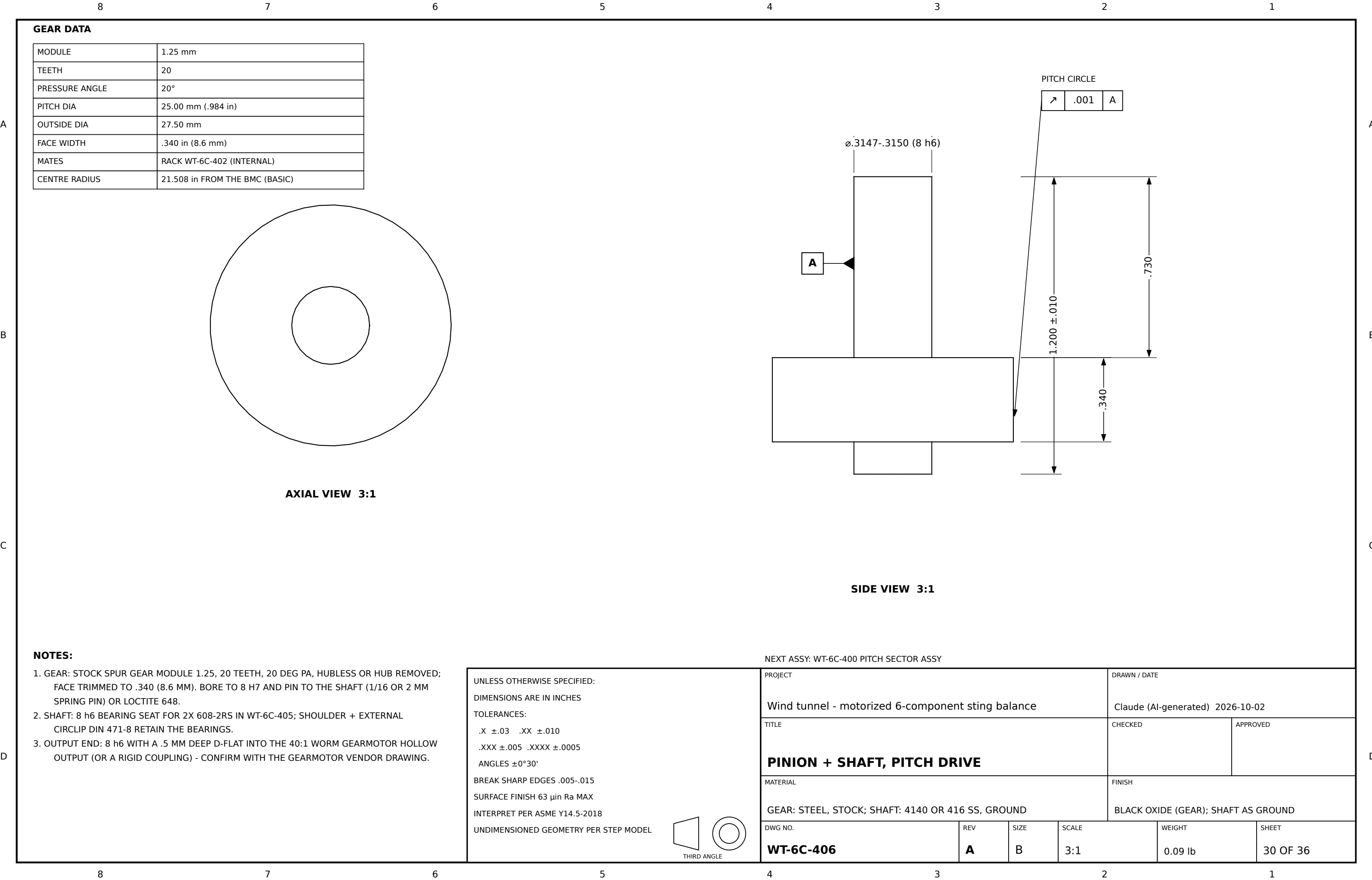
UNDIMENSIONED GEOMETRY PER STEP MODEL

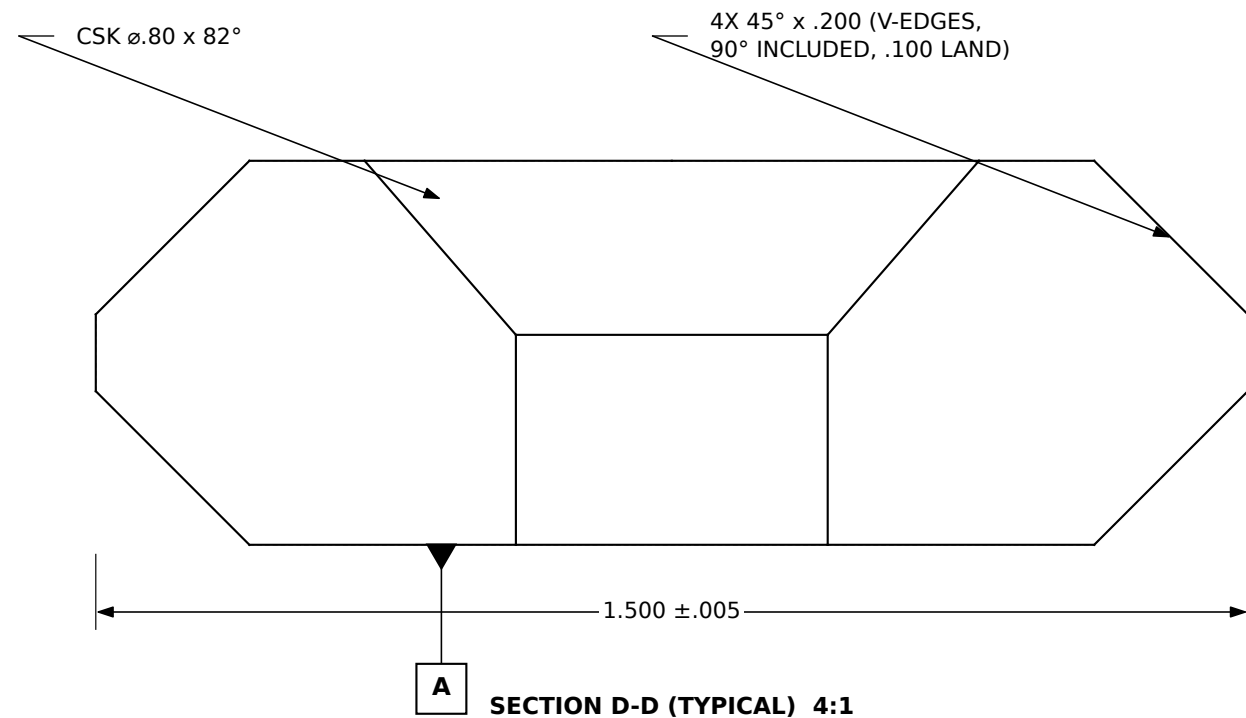


THIRD ANGLE

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
RACK PLATE, PITCH (INTERNAL ARC RACK)								
MATERIAL				FINISH				
6061-T6 ALUMINUM, .500 PLATE				CLEAR ANODIZE TYPE II (TEETH INCLUDED)				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-402		A	B	1:2.2 / 3:1		0.89 lb		26 OF 36





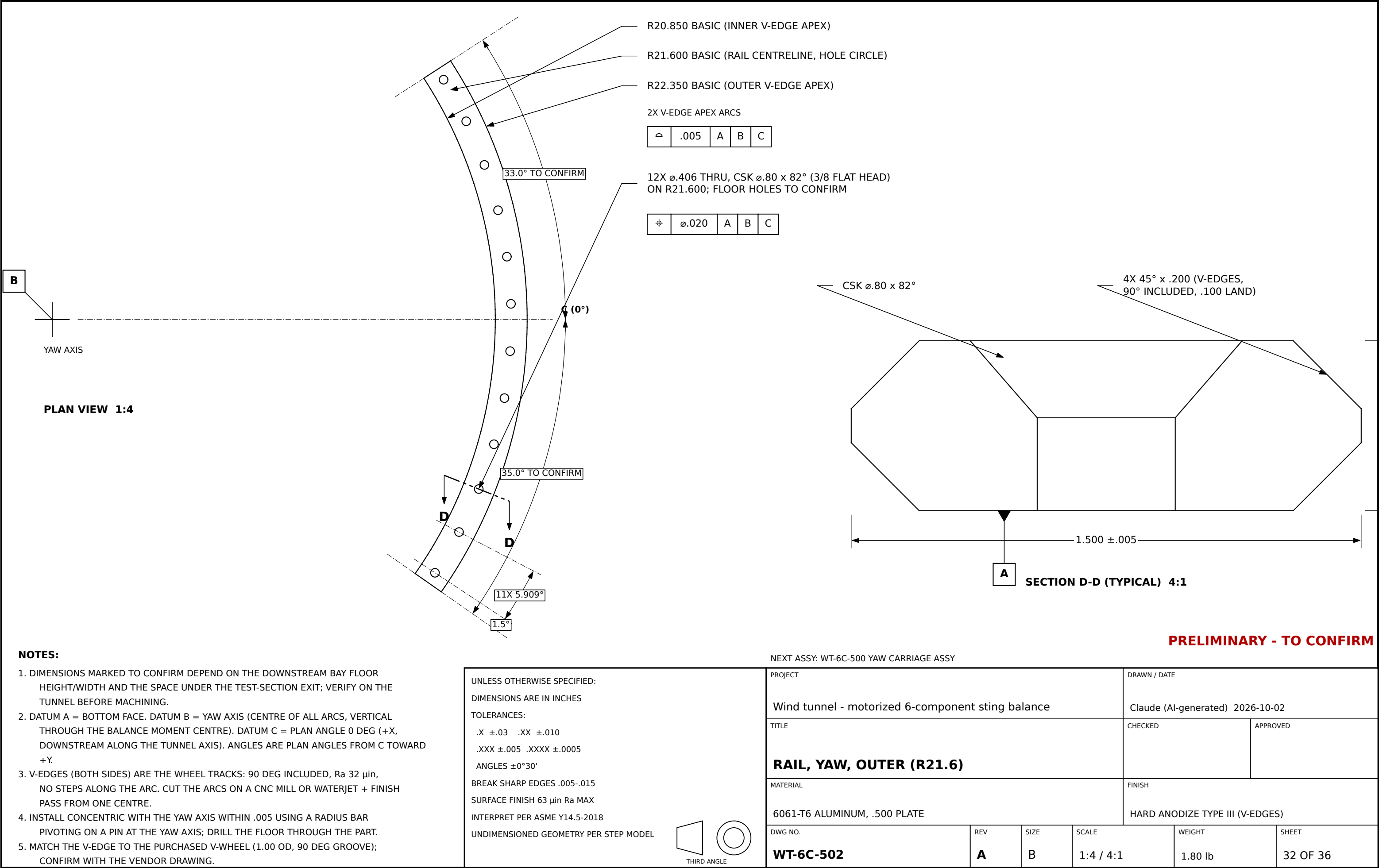
1. DIMENSIONS MARKED TO CONFIRM DEPEND ON THE DOWNSTREAM BAY FLOOR HEIGHT/WIDTH AND THE SPACE UNDER THE TEST-SECTION EXIT; VERIFY ON THE TUNNEL BEFORE MACHINING.
2. DATUM A = BOTTOM FACE. DATUM B = YAW AXIS (CENTRE OF ALL ARCS, VERTICAL THROUGH THE BALANCE MOMENT CENTRE). DATUM C = PLAN ANGLE 0 DEG (+X, DOWNSTREAM ALONG THE TUNNEL AXIS). ANGLES ARE PLAN ANGLES FROM C TOWARD +Y.
3. V-EDGES (BOTH SIDES) ARE THE WHEEL TRACKS: 90 DEG INCLUDED, Ra 32 μ in, NO STEPS ALONG THE ARC. CUT THE ARCS ON A CNC MILL OR WATERJET + FINISH PASS FROM ONE CENTRE.
4. INSTALL CONCENTRIC WITH THE YAW AXIS WITHIN .005 USING A RADIUS BAR PIVOTING ON A PIN AT THE YAW AXIS; DRILL THE FLOOR THROUGH THE PART.
5. MATCH THE V-EDGE TO THE PURCHASED V-WHEEL (1.00 OD, 90 DEG GROOVE); CONFIRM WITH THE VENDOR DRAWING.

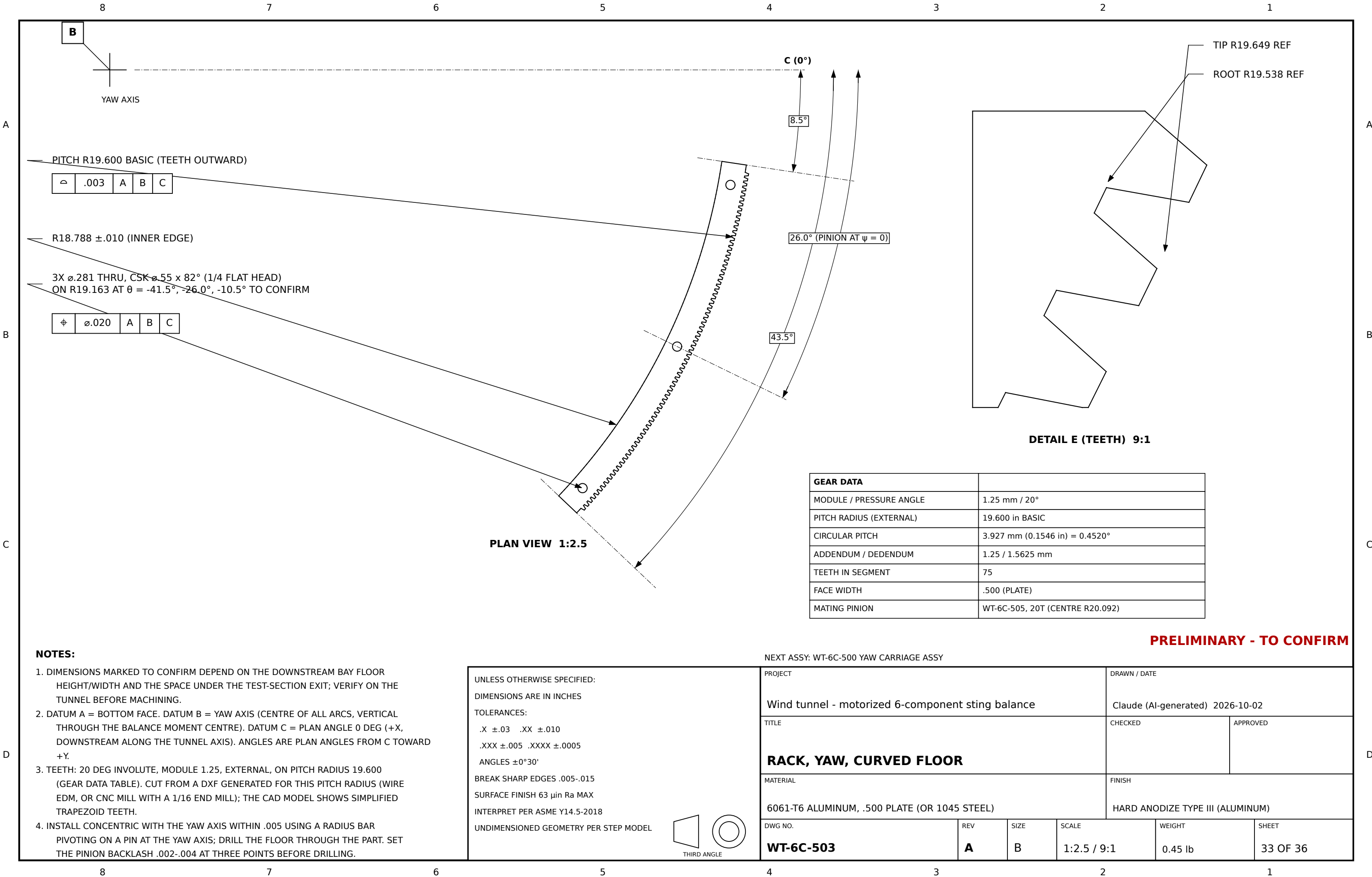


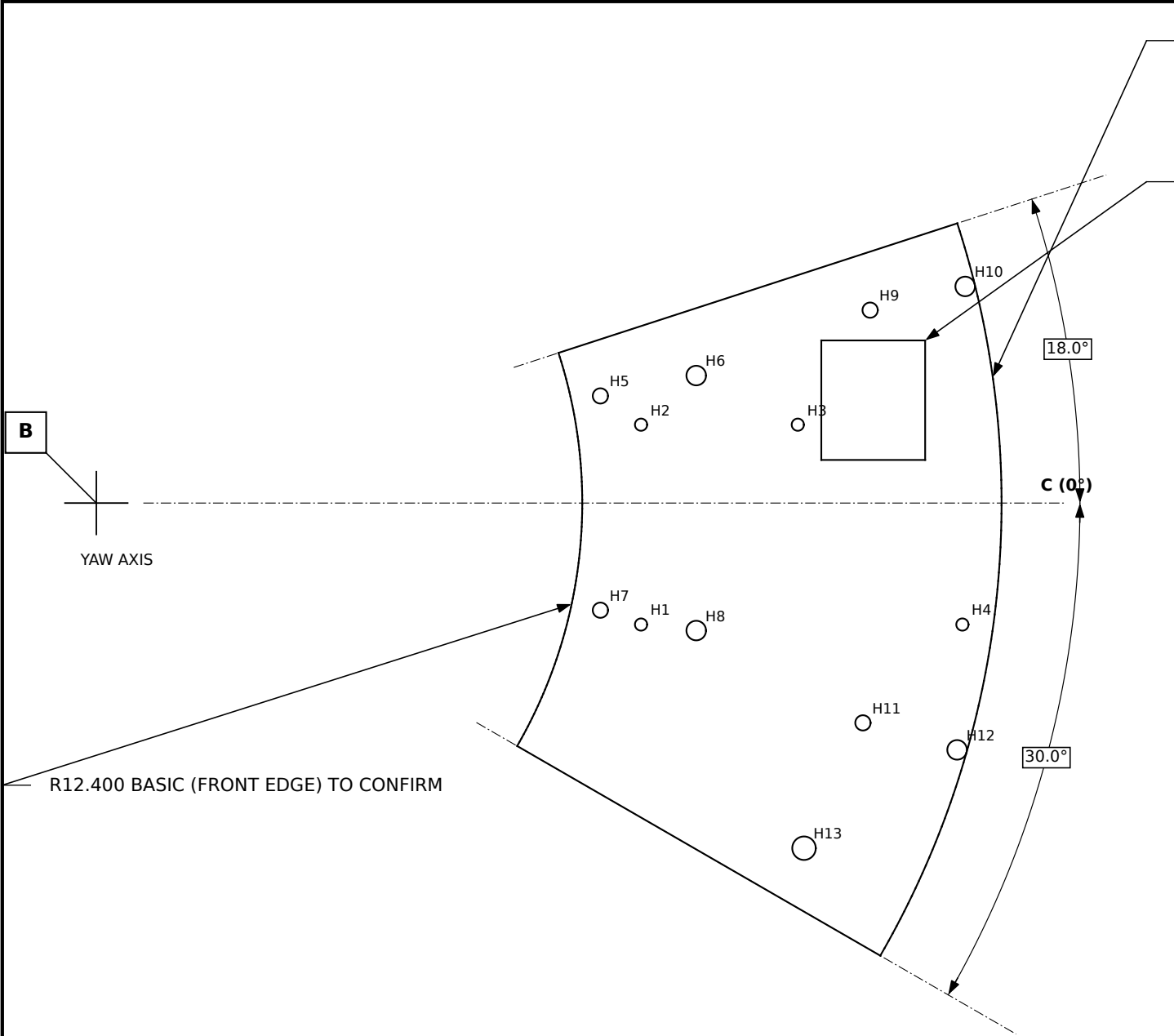
THIRD ANGLE

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
RAIL, YAW, INNER (R14.4)							
MATERIAL				FINISH			
6061-T6 ALUMINUM, .500 PLATE				HARD ANODIZE TYPE III (V-EDGES)			
DWG NO.		REV	SIZE	SCALE	WEIGHT		SHEET
WT-6C-501		A	B	1:3 / 4:1	1.07 lb		31 OF 36

PRELIMINARY - TO CONFIRM







R23.100 BASIC

OUTLINE

	.020	A	B	C
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NOTCH THRU: X 18.50-21.15, Y 1.10-4.15 BASIC
(CLEARS THE ROLL NEMA 23 AT +20° PITCH)

DATUM A: TOP FACE

	.005
--	------

HOLE TABLE (BASIC FROM B / C) - POSITION $\varnothing$.010 A B C

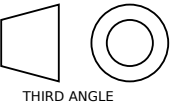
HOLE	SIZE	X	Y	R	θ	FOR
H1	3/8-16 UNC-2B THRU	13.900	-3.100	-	-	SECTOR BASE
H2	3/8-16 UNC-2B THRU	13.900	+2.000	-	-	SECTOR BASE
H3	3/8-16 UNC-2B THRU	17.900	+2.000	-	-	SECTOR BASE
H4	3/8-16 UNC-2B THRU	22.100	-3.100	-	-	SECTOR BASE
H5	$\varnothing$.390 THRU	12.863	+2.734	13.150 TO CONFIRM	+12.0°	WHEEL STUD
H6	$\varnothing$.500 THRU	15.308	+3.254	15.650 TO CONFIRM	+12.0°	ECC. SPACER
H7	$\varnothing$.390 THRU	12.863	-2.734	13.150 TO CONFIRM	-12.0°	WHEEL STUD
H8	$\varnothing$.500 THRU	15.308	-3.254	15.650 TO CONFIRM	-12.0°	ECC. SPACER
H9	$\varnothing$.390 THRU	19.746	+4.923	20.350 TO CONFIRM	+14.0°	WHEEL STUD
H10	$\varnothing$.500 THRU	22.171	+5.528	22.850 TO CONFIRM	+14.0°	ECC. SPACER
H11	$\varnothing$.390 THRU	19.562	-5.609	20.350 TO CONFIRM	-16.0°	WHEEL STUD
H12	$\varnothing$.500 THRU	21.965	-6.298	22.850 TO CONFIRM	-16.0°	ECC. SPACER
H13	$\varnothing$.600 THRU	18.059	-8.808	20.092	-26.0°	PINION SHAFT

PLAN VIEW (TOP) 1:4

NOTES:

- DIMENSIONS MARKED TO CONFIRM DEPEND ON THE DOWNSTREAM BAY FLOOR HEIGHT/WIDTH AND THE SPACE UNDER THE TEST-SECTION EXIT; VERIFY ON THE TUNNEL BEFORE MACHINING.
- DATUM A = TOP FACE (SECTOR BASE SEATS HERE), FLAT .005. DATUM B = YAW AXIS (CENTRE OF THE ARCS). DATUM C = PLAN ANGLE 0 (+X DOWNSTREAM). HOLE TABLE IN BASIC COORDINATES FROM B ALONG C.
- WHEEL STUD HOLE RADII ASSUME A 1.00 OD WHEEL TANGENT TO THE RAIL V-EDGE APEX: RESET THEM FROM THE PURCHASED WHEEL GROOVE DEPTH (TO CONFIRM). OUTER HOLE OF EACH TRUCK IS .500 FOR THE ECCENTRIC SPACER (WT-6C-506-2).
- FRONT (INNER) EDGE R12.400 PASSES UNDER THE TEST-SECTION EXIT AT +-15 DEG YAW ($x = 10.7$ IN).
- LIGHTENING POCKETS AFTER FEA (NOT SHOWN).

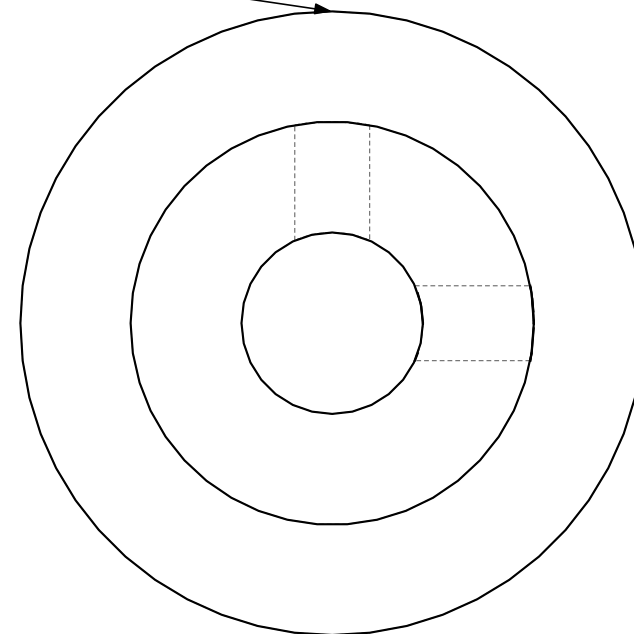
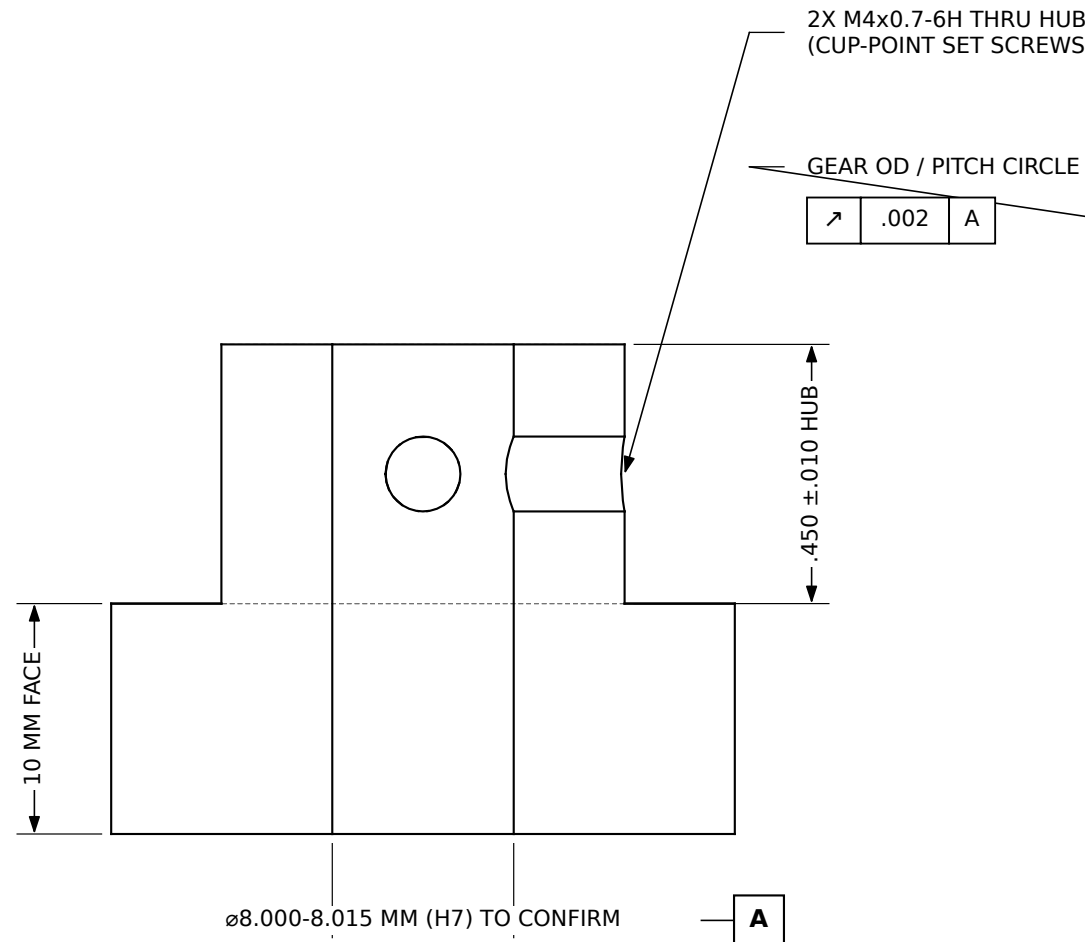
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X $\pm$.03 .XX $\pm$.010
.XXX $\pm$.005 .XXXX $\pm$.0005
ANGLES $\pm$ 0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 μ in Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL



NEXT ASSY: WT-6C-500 YAW CARRIAGE ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
PLATE, YAW CARRIAGE								
MATERIAL				FINISH				
6061-T6 ALUMINUM, .500 PLATE (MIC-6 PREFERRED)				CLEAR ANODIZE TYPE II				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-504		A	B	1:4		7.27 lb		34 OF 36

PRELIMINARY - TO CONFIRM



GEAR DATA	
MODULE / PA	1.25 mm / 20°
TEETH	20
PITCH DIA	25.000 mm (.984 in)
OUTSIDE DIA	27.500 mm (1.083 in)
FACE	10 mm
MATES	WT-6C-503 RACK

SECTION A-A 3:1

TOP VIEW 3:1

NOTES:

1. PURCHASE A STOCK m1.25 x 20T x 10 MM FACE STEEL SPUR GEAR WITH HUB (e.g. KHK SS1.25-20 OR EQUIVALENT). MODIFY ONLY: BORE, SET SCREWS, HUB LENGTH
2. BORE SIZE AND KEY / FLAT TO SUIT THE 30:1 WORM GEARMOTOR OUTPUT SHAFT (VENDOR DRAWING, TO CONFIRM). 8 MM SHOWN.
3. THE GEAR RUNS AT RAIL LEVEL, 1.45 IN BELOW THE CARRIAGE PLATE TOP: IF THE GEARMOTOR SHAFT IS SHORTER, ADD A SHAFT EXTENSION + RIGID COUPLING (TO CONFIRM).
4. MESH WITH THE FLOOR RACK WT-6C-503 AT CENTRE RADIUS 20.092; BACKLASH .002-.004.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03	.XX ±.010
---------	-----------

.XXX ±.005 .XXXX ±.0005

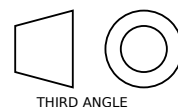
ANGLES $\pm 0^{\circ}30'$

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 $\mu\text{in Ra MAX}$

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



NEXT ASSY: WT-6C-500 YAW CARRIAGE ASSY

PROJECT				DRAWN / DATE		
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02		
TITLE				CHECKED		APPROVED
PINION, YAW (MODIFIED STOCK GEAR)						
MATERIAL				FINISH		
STEEL SPUR GEAR, m1.25 20T 20° PA, WITH HUB				AS SUPPLIED; BLACK OXIDE		
DWG NO.		REV	SIZE	SCALE	WEIGHT	SHEET
WT-6C-505		A	B	3:1	0.13 lb	35 OF 36

PRELIMINARY - TO CONFIRM

